INTERNATIONAL MONETARY FUND

IMFNow: The IMF Nowcasting Framework in R

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WP/26/240

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2026
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WORKING PAPER

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IMF Working Paper
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IMFNow: The IMF Nowcasting Framework in R

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Authorized for distribution by Axel Schimmelpfennig
October 2026

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ABSTRACT: Nowcasting GDP in low-income and fragile economies poses a set of problems that bind at once: short and noisy samples, ragged-edge releases, non-stationary regressors, model-selection uncertainty, the need to communicate uncertainty and attribution to non-technical audiences, and frequent within-quarter updates. Existing nowcasting toolkits, designed largely for data-rich economies, do not address this combination. This paper presents **IMFNow**, the open-source R package implementing the International Monetary Fund's nowcasting framework and how it addresses these constraints. In one reproducible pipeline, it integrates sixteen machine-learning and time-series learners, circular block-bootstrap cross-validation for short dependent samples, automatic stationarity handling with analyst override, expanding-window conformal prediction intervals calibrated to propagate model-selection uncertainty, the Model Confidence Set based on out-of-sample losses, SHAP decompositions, and a fast-update mode that reuses tuned hyperparameters within a quarter. The framework is delivered through two functions, **macroNowcast** and **outputNowcast**. We motivate each component by the forecasting problem it solves, document the architecture, and demonstrate the framework on quarterly GDP growth for Uganda and Tunisia using rolling-origin pseudo-real-time evaluation of point accuracy and interval coverage.

RECOMMENDED CITATION: Abdel-Latif, H. & Centorrino, S.& Dovi, M.S, (2026). "IMFNow: The IMF Nowcasting Framework in R." IMF Working Paper No. WP/26/240.

JEL Classification Numbers:	C45, C53, C55, C87, E37
Keywords:	IMFNow, nowcasting, machine learning, time series, conformal prediction
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We are grateful to Christian Ebeke, Natan Epstein, Andreas Hake, Xingwei Hu, Margarida Martins, Borislava Mircheva, Juan Prada Sarmiento, Helen Qiaoluan, Srikanth Ramamurthy, and Kadir Muhittin Tanyeri for helpful comments and suggestions. Any remaining errors are our own.

WORKING PAPERS

IMFNow: The IMF Nowcasting Framework in R

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1. Introduction

Economic policymakers and analysts increasingly rely on nowcasting, the prediction of present or very near-term economic conditions, to support timely decision-making. Traditional macroeconomic indicators, such as GDP growth, are typically released with substantial lags, often several weeks after the reference period and sometimes months in low-income and fragile economies. This information gap can be especially costly in times of rapid change, including financial crises, commodity price shocks, and global health emergencies. To bridge it, institutions such as the International Monetary Fund (IMF) integrate high-frequency indicators (industrial production, retail sales, energy consumption, financial market prices, and satellite-based measures of activity) into models that produce real-time estimates of economic conditions. During the COVID-19 pandemic, for example, nowcasts that incorporated mobility data and electricity usage delivered early warnings of sharp contractions well ahead of official GDP releases. Barhoumi et al. (2022) document how machine-learning-based nowcasting captured the drop in Botswana's economic activity weeks before the official statistics were available, building on earlier IMF work using machine learning to nowcast economies with limited data infrastructure (Tiffin, 2016).

Most of the existing nowcasting literature, and the software that supports it, has been developed in advanced economies, where samples are longer, data releases are more predictable, and analyst expertise is concentrated. The dominant tradition combines diffusion-index and dynamic-factor models with mixed-frequency information (Stock and Watson, 2002; Giannone et al., 2008; Bańbura et al., 2011, 2013, among others), and the more recent big-data and machine-learning nowcasting systems have largely been built for the same data-rich environments (Bok et al., 2018; Dauphin et al., 2022). Machine-learning methods can deliver competitive performance for macroeconomic forecasting once the curse of dimensionality is controlled (Stock and Watson, 2012; Medeiros et al., 2021; Goulet Coulombe et al., 2022). The setting we have in mind is different. The motivating use case for `IMFNow` is the regular surveillance of low-income and fragile economies inside the IMF, where the constraints are stark and largely shared across countries.

Samples are short, with many country panels containing only a few dozen quarterly observations after harmonization, often fewer than the cyclical horizon of interest. They are also ragged at the edge, because indicators arrive at different lags and the most recent quarter is almost always only partially observed. The available indicators are recorded at mixed frequencies, with survey, financial, and satellite series sampled monthly, weekly, or daily against a quarterly target; the package meets this through its ordinary regressor interface, either by aggregating higher-frequency series to the target's frequency or by entering their individual high-frequency lags as regressors in the unrestricted-MIDAS manner (see Section 3).[1] Finally,

[1] See Ghysels et al. (2004, 2007) and Foroni et al. (2015) on MIDAS and unrestricted MIDAS regressions, Clements and Galvão (2008), Andreou et al. (2010), and Andreou et al. (2013) on mixed-frequency regression and forecasting, and Marcellino and Schumacher (2010) and Foroni and Marcellino (2014) on nowcasting with ragged-edge, mixed-frequency data.

regressors are frequently non-stationary. Levels and trends are common, and the same predictor may be stationary in one country and integrated in another.

Beyond these data problems, model-selection uncertainty is severe. With so few observations no single model dominates, and the cross-validation winner is itself a noisy estimate. Uncertainty must be communicated plainly to a policy audience that is typically a desk economist or a senior official rather than a statistician, which calls for intervals that are calibrated, distribution-free, and comparable across models. Interpretability is equally at a premium, because the non-linear ensembles that tend to win on out-of-sample accuracy are not legible to an analyst preparing a country briefing, so attributions for non-technical readers are part of the deliverable. And nowcasts are subject to an operational refresh constraint: they are produced not once but every time a relevant indicator is released, sometimes weekly within a quarter, which rules out re-tuning the full hyperparameter grid on every update.

These constraints shape every component of the `IMFNow` package. Rather than wrap a single forecasting model in a user-friendly interface, `IMFNow` is built around the question of what a nowcasting workflow needs to look like when each of these constraints binds at once. Its answer is a framework in which each constraint is met by a specific component, and the components are made to work together: block-bootstrap cross-validation designed for hyperparameter tuning on short dependent samples, automatic stationarity handling with an analyst override, expanding-window conformal calibration that carries model-selection uncertainty into the prediction intervals, a Model Confidence Set computed on the same out-of-sample predictions, SHAP decomposition of every nowcast, and a refresh mode that reuses the tuned configurations. The whole pipeline sits behind two top-level functions, `macroNowcast` and `outputNowcast`, so that it runs the same way on every country desk and every data vintage. The intended users are economists who produce nowcasts in data-scarce settings, at the IMF's nowcasting hub (NowHub) and country desks, in central banks, finance ministries and statistical offices, and in applied research.[2] They assemble the predictor set from a country's release calendar, run the two functions when new data arrive, and carry the resulting point nowcast, interval, model ranking, and attribution into a briefing or a paper. The package automates estimation, calibration, and reporting, while specification (e.g., which indicators, transformations, and sample) remains with the analyst, a division of labour we return to in Section 2.8.

The paper makes four contributions. First, it identifies the set of forecasting problems that jointly define nowcasting in low-income and fragile economies and derives from them the design of a complete nowcasting framework, rather than treating each problem in isolation as the literature has largely done. Second, it provides the open-source implementation of that framework, `IMFNow`, which is the tool NowHub uses for country surveillance: a reproducible, seeded pipeline with a two-function interface, a suite of sixteen learners

[2] `IMFNow` is designed with these settings in mind, but it can equally be used in data-rich settings.

that users can subset or extend, and uncertainty and attribution outputs designed for policy communication.[3] Third, it sets out the methodology as implemented, in enough detail to be audited and reproduced: the block-bootstrap cross-validation, the expanding-window conformal calibration with multiplicative scaling, the out-of-sample Model Confidence Set, and the treatment of non-stationarity and ragged edges. Fourth, it demonstrates the framework end-to-end on quarterly GDP nowcasts for Uganda and Tunisia and evaluates it in a rolling-origin pseudo-real-time exercise that reports out-of-sample accuracy against a univariate benchmark and the empirical coverage of the conformal intervals, including the diagnostics that tell an analyst when a result should be taken with caution. The same trade-offs arise in adjacent data-scarce surveillance settings, and the design transfers directly. The finite-sample properties of the methodology are the subject of a companion paper (Abdel-Latif et al., 2026), which studies the block-bootstrap cross-validation and the conformal calibration by Monte Carlo simulation at the sample sizes of this setting and compares the models across a panel of about sixty economies; we cite its findings where they bear on a design choice.

The remainder of the paper is organized as follows. Section 2 discusses the forecasting problems above and the design responses they motivate. Section 3 describes the resulting software architecture and the user-facing interface. Section 4 walks through a full nowcasting example for Uganda and uses it to illustrate the design choices in action, including forecasting beyond the last available observation. Section 5 reports a compact second application to Tunisia, Section 6 presents the pseudo-real-time evaluation, Section 7 turns to use on a desk: prediction intervals at multiple confidence levels and the operational fast-update mode. Section 8 concludes. The appendices give the algorithms as implemented (block-bootstrap cross-validation, conformal calibration), the hyperparameter search space, the predictor lists, and the detailed evaluation tables. This paper is based on version 0.0.7.1 of `IMFNow`.

2. From forecasting problems to design responses

We take each of the forecasting problems listed above in turn, explain why off-the-shelf practice is not adequate in our setting, and describe the design response we adopted in `IMFNow`. The user-facing surface in Section 3 and the example in Section 4 follow directly from these choices.

2.1. Short samples and the overfitting problem

In the typical IMF country panel, the history of quarterly real GDP growth, especially in low-income countries (LICs), runs only from the early 2010s to the most recent release, leaving around fifty observations or fewer

[3] While NowHub estimates may inform surveillance, they do not replace official IMF forecasts or official statistics in any way or form, nor do they represent official IMF forecasts. They are estimates that move frequently with data releases and are meant to support policy discussion.

after differencing and lag construction. With even a moderately rich predictor set, hold-out validation on a single end-of-sample test window is too noisy to discriminate among models, and standard k-fold cross-validation breaks the temporal ordering. Both extremes are common pitfalls: the first picks the model that happened to fit the last recession; the second picks the model that interpolates best across time, which is not what we want. This overfitting risk is borne out empirically on short GDP series, where regularized linear learners tend to outperform more complex non-linear learners that overfit the limited history (Akepanidtaworn and Akepanidtaworn, 2025).

We adopt a *circular block-bootstrap cross-validation* scheme, implemented in `cv_plan_custom` and described step by step in AppendixA. Overlapping blocks of length b are constructed, one starting at each observation, wrapping around the end of the series in the spirit of Künsch (1989), Politis and Romano (1994), and Bühlmann (2002). For each replication, a contiguous assessment window of up to four observations is held out, every block that overlaps with that window is excluded from the training pool to prevent leakage, and the remaining blocks are sampled with replacement to form the training set. The block length is selected data-adaptively from the target series with the circular-bootstrap estimator of Patton et al. (2009), with a floor of four observations and a cap of one quarter of the sample. The user faces this entirely through the `cv_param` argument of `macroNowcast`: a small named list of two integers, `reps` and `block_size`, with a documented default that requires no tuning in routine use. The scheme is the methodological contribution of Abdel-Latif et al. (2026), who study it by simulation and finds that, at fifty observations, the models it selects and the inverse-loss ensemble improve on naive benchmarks across sparse, dense and factor designs.

The circular construction deserves a comment, because wrapping ties the end of the sample to its beginning. Only the $b-1$ blocks that start in the last $b-1$ positions wrap, so a share $(b-1)/n$ of the draws contains a single artificial transition from the last observation to the first (about one in four for Uganda's block length of 13 and one in seventeen for Tunisia's block length of 4, AppendixA); the held-out assessment window itself wraps with probability of order b/n. We accept this cost deliberately: a non-circular scheme under-samples the end of the sample, and the most recent regime is precisely the one being nowcast. A user who considers the wrap harmful for a series with a pronounced level shift can shorten `block_size`, which reduces the share of wrapped draws proportionally.

2.2. Ragged edges and the trailing-NA convention

The second problem is the most recent quarter, for which a few high-frequency indicators are already available while GDP growth is not. We adopt a *trailing-NA convention*: the row to be nowcast is the row whose dependent variable is missing. The user leaves the trailing values of the target as `NA`, retains whatever regressors are available for those rows, and passes the data frame to `macroNowcast`, which holds those rows aside as the prediction set and returns one nowcast per missing row. There is no separate "horizon"

argument and no parallel "forecast data" object; the same data frame describes both the training history and the rows to be predicted, and shifting the target by an extra horizon (Section 4.6) is a simple `lead()` away.

A second, distinct manifestation of ragged-edge data lives in the *predictor* columns rather than the target. In a realistic publication cycle, the trailing quarter typically has the exchange rate and the satellite indicators already available, while a survey-based index for the same quarter has not been released. Treating such a row as if all predictors were available would discard it from the prediction set; treating it as a stop-the-pipeline event would force the user to impute upstream by hand, which is exactly the kind of operational friction the package is designed to remove. We expose this case through the `na_handling` argument of `macroNowcast`, which dispatches between five rules. The default, `"impute_linear"`, imputes each incomplete predictor by a linear regression on *all other predictors* (a multivariate imputation, not a univariate one), estimated inside the `tidymodels` recipe of every non-ARIMAX engine via `recipes::step_impute_linear` and, for the ARIMAX engine's exogenous-regressor matrix, by the package's `impute_predictors_linear()` helper. Because the recipe is re-estimated inside every cross-validation resample and every calibration window, the imputation never sees the assessment observations. The alternatives are `"forward_fill"` (carry the last observed value forward), `"drop_predictor"`, `"drop_row"` (the historical behaviour, retained for users who prefer it), and `"require_complete"` (which errors).

Two consequences of this design should be emphasised. First, the prediction interval at the nowcast row is conditional on the imputed predictor values: historical rows almost never have missing predictors, so the conformity scores that calibrate the interval do not include imputation error, and the band does not widen when a predictor is imputed. Second, imputation is the right tool for *occasional* gaps, not for *systematic* publication lags. For an indicator that is always released after the GDP nowcast is due, the better construction is to train the model on the lag of that indicator (a user-built lagged column, or `lags_x`), which is the unrestricted-MIDAS practice of using the information actually available at the nowcast date; `"forward_fill"` is the natural choice for slow-moving indicators. Section 4.5 quantifies, for the Uganda example, how much the nowcast moves under each treatment when one predictor is missing in the nowcast quarter. A row-level diagnostic is always produced: when any row intended for nowcasting has a missing predictor, the function names the affected dates and predictors in a warning so the user is never silently denied a nowcast.

2.3. Non-stationary regressors and automatic differencing

A recurring source of bugs in applied nowcasting is silent non-stationarity. The same indicator (say, CPI) may be stationary in one country and integrated in another; manual differencing decisions made on one country tend to leak into another via copy-paste; and back-transformation, when forgotten, produces level forecasts that diverge from the original units of the target.

We make stationarity testing and back-transformation a built-in step of `macroNowcast`. With the default

`difference = "auto"`, the dependent variable and each predictor are tested for a unit root using the KPSS test (Kwiatkowski et al., 1992) embedded in `forecast::auto.arima` (Hyndman et al., 2024; Hyndman and Khandakar, 2008). If the test indicates a unit root, the series is first-differenced before model fitting. For the dependent variable, the nowcast is produced in differences and then *automatically* back-transformed to levels by adding the lagged level, both for the point forecast and for every interval bound, at every confidence level. The user always receives results in the original units of the target, and the decision taken for every series is returned in the `differencing` element of the fitted object (and printed, for the two examples, in AppendixD).

The automatic decision can be overridden. `difference = "none"` switches the test and the differencing off; a character vector of variable names differences exactly those series without testing; and the `differencing` table of a previous fit replays that fit's decisions, which is what the fast-update mode of Section 7.2 should use so that a refresh applies the same transformation as the tuning run. The override matters because the KPSS test has known finite-sample size distortions (Caner and Kilian, 2001): at the sample sizes typical of LIC surveillance (dozens of quarters rather than hundreds) it can over-reject stationarity and difference a series that is in fact trend-stationary. The costs of the two possible errors are asymmetric. Because stationarity is the null hypothesis, over-rejection errs towards differencing, which costs one observation and some low-frequency signal but does not invalidate the fit; failing to difference an integrated predictor does. We therefore keep the automatic rule as the default and give the analyst the override rather than combining the KPSS with tests of the opposite null (ADF, ERS), whose concordance rule would have no defined size and would push the decision towards the costlier error.

Structural breaks are a related concern: a level shift (a regime change, a policy reform, an exchange-rate realignment) can masquerade as a unit root (Perron, 1989) and drive a spurious differencing decision, or alter the nowcasting relationship itself. We deliberately do not automate break detection inside `macroNowcast`. At fifty observations the tests have little power, so an automatic flag would be noisy, and there is no principled automatic remedy: truncating the sample after a break discards the data the models need, and dummying a break requires judgement about its nature. We instead recommend that users running the package on a new country pair an initial diagnostic run with break checks on the dependent variable and the main predictors, for example the supremum-*F* and CUSUM tests of `strucchange` (Zeileis et al., 2002) or the multiple-breakpoint procedure of Bai and Perron (2003) (a five-line snippet is given in AppendixF), and act on the result through the `difference` override, the choice of sample, or an explicit break regressor. The non-linear learners in the suite can accommodate level shifts that are reflected in the predictors; what they cannot do is detect a break in a relationship that the data do not yet contain.

2.4. Model-selection uncertainty in small samples

In a short sample, the cross-validation winner is itself a noisy estimate: re-tuning on a slightly different data vintage routinely produces a different "best" model. Reporting the winner alone is therefore misleading, both in point and in interval form, because it ignores the variance contributed by the selection step.

`IMFNow` responds at three levels. First, alongside the best individual model, the package reports an *ensemble nowcast* constructed as a weighted average of all candidate models, with weights inversely proportional to each model's average cross-validated loss for accuracy-style metrics (RMSE, MAE) and proportional to the metric for concordance-style metrics (CCC, R^2). Models that perform poorly on the cross-validation sample receive smaller weights but are not discarded, an approach that has long been recommended for short macroeconomic series (Timmermann, 2006). We choose inverse-loss weighting over alternatives such as ordinary least-squares-optimal weights (Bates and Granger, 1969; Granger and Ramanathan, 1984), Bayesian model averaging, or simple equal weighting because optimal weights estimated from a few dozen observations are noisy enough that their weight uncertainty routinely swamps the diversification gain. This is the *forecast combination puzzle* documented by Stock and Watson (2004) and explained by Smith and Wallis (2009) as a small-sample bias-variance trade-off; in repeated comparisons of professional and model forecasts, the simple average and other low-complexity schemes are difficult to beat (Genre et al., 2013). Inverse-loss weighting preserves the diversification benefit while avoiding the weight-estimation noise that derails fully data-driven combinations in our setting.[4]

Second, the package wraps the Model Confidence Set (MCS) procedure of Hansen et al. (2011), via the `MCS` package (Catania and Bernardi, 2017), to return the subset of models that contains the best model with a given level of confidence: a confidence set for *which* model to use, in the spirit of the stepwise multiple-testing framework of Romano and Wolf (2005) and extending the pairwise predictive-accuracy testing of Diebold and Mariano (1995) to the multiple-model case. The procedure is sequential. Starting from the full candidate set (all models plus the ensemble), it tests the null hypothesis of equal predictive ability across the current set; if the null is rejected, the model with the worst relative performance is eliminated and the test is repeated on the survivors; it stops at the first non-rejection, and the surviving set is the MCS. Each model is reported with two p-values: the p-value of the equal-predictive-ability test at the step in which that model was (or would have been) eliminated, and the *MCS p-value*, the running maximum of the step p-values up to that model. A model belongs to the $(1 - \alpha)$ MCS exactly when its MCS p-value is at least α. The loss matrix is built from genuinely out-of-sample predictions: the squared errors of every model, and of the ensemble, at the expanding-window calibration dates described below, where each model is refitted on data strictly

[4] Information-criterion weights (AIC, BIC) are not an option in this suite: the criteria are undefined for most of the learners (random forests, boosted trees, BART, Cubist, neural networks) and not comparable across model families, and they are in-sample quantities, whereas the cross-validated losses that drive our weights are computed on held-out blocks.

before the date it predicts. The use of the historical fitted values of the full-sample models, which rewards flexible learners that fit the history closely, is retained only as a labelled fallback for objects that carry no out-of-sample predictions.

Third, prediction intervals are constructed in the distribution-free conformal framework (Lei et al., 2018; Angelopoulos and Bates, 2023) by *expanding-window (leave-future-out) conformal calibration*: each model's selected configuration is refitted on every expanding window of the history, its error on the next observation is recorded, and the interval is the point nowcast plus or minus a finite-sample quantile of those errors. For the ensemble, the errors are studentised by the dispersion of a pseudo-population of candidate predictions (the top-K hyperparameter configurations of every model) before the quantile is taken, which is the multiplicative PCS-UQ calibration of Agarwal et al. (2025): the interval widens where the candidate models disagree and narrows where they agree, so the selection uncertainty is propagated into the reported bounds. The closest relatives of this construction are the jackknife+ of Barber et al. (2021), from which it borrows the refit-and-score logic but not the aggregation of leave-one-out predictions, and the ensemble batch prediction intervals of Xu and Xie (2021), which also calibrate on sequential out-of-sample residuals. The procedure is described exactly as implemented in AppendixB. A faster split-conformal alternative, calibrated on the second temporal half of the sample, is also available.

The user surface for these three responses is intentionally minimal: `outputNowcast` returns the ensemble nowcast and the MCS in the same object as the best-model nowcast (`MCS_results`, `MCS_survivors`, plus an `Ensemble` row in `all_models_nowcast`), and the conformal calibration mode is controlled by the single argument `conf_method`.

2.5. Honest uncertainty for policy audiences

Even when the point forecast is accurate, a nowcast that arrives with a single number is hard to use. Policy audiences (briefing notes, internal review meetings, central bank publications) expect a range, often presented as a fan chart with several confidence levels. Standard parametric prediction intervals from individual models are not adequate here: they are calibrated under model-specific assumptions that fail in short, possibly non-normal samples, and they are not directly comparable across models.

We compute prediction intervals at four standard nominal levels (80%, 90%, 95%, 99%) for *every* model in the suite, and for the ensemble, in a single pass, using the same conformal procedure across all of them; the conformity scores are computed once and only the quantile step differs by level. This is exposed via the `all_models_nowcast` data frame returned by `outputNowcast`, which includes a `conf_level` column so that fan charts can be assembled directly with `dplyr::filter` (illustrated in Section 7.1). One consequence of the finite-sample construction must be understood by users of short samples: with n conformity scores the $(1-\alpha)$ bound is the $\lceil (1-\alpha)(n+1) \rceil$-th smallest score, and when that index exceeds n the bound is infinite.

With fewer than 99 calibration dates, which is every quarterly panel in LIC surveillance, the 99% interval is therefore unbounded, and the package reports it as such rather than extrapolating. The 95% interval requires at least 19 scores and the 90% interval at least 9.

We adopt expanding-window calibration because, in the short samples typical of LIC surveillance, it makes efficient use of every observation while remaining distribution-free and model-agnostic. We acknowledge that it leans on a stationarity-type assumption on the errors, so that past errors are informative about the next one, which the non-exchangeable conformal framework relaxes directly by reweighting nonconformity scores to downweight distant observations (Barber et al., 2023), and that strict time-series conformal methods relax through ergodicity-based arguments (Chernozhukov et al., 2018), ensemble-based interval construction over rolling windows (Xu and Xie, 2021), or online adaptive recalibration (Zaffran et al., 2022). These alternatives become preferable when the sample is long enough to absorb their data cost, and we view their integration into the package as an extension rather than a methodological revision. The empirical coverage of the intervals as implemented is reported in Section 6.[5]

2.6. Black-box interpretability for non-technical readers

The package's default suite includes models with very different inductive biases, including non-linear ensembles such as random forests, XGBoost, and BART. These models often deliver the best out-of-sample performance, but their internal representation is not interpretable to a desk economist preparing a country briefing.

We make SHAP decomposition a first-class output of every nowcast. Inside `macroNowcast`, the best model is wrapped with `DALEXtra::explain_tidymodels`, and the SHAP values for the last missing quarter are computed via `DALEX::predict_parts`, which implements the sampling-based Shapley-value estimator of Štrumbelj and Kononenko (2014) with `shap_B` Monte Carlo permutations (default 100, seeded by the run). We chose this model-agnostic estimator over family-specific accelerations such as TreeSHAP (Lundberg et al., 2020) so that exactly the same decomposition machinery applies to the linear, tree-based, neural-network, and time-series engines in the suite. The cost is an $O(B \cdot J)$ overhead per explained observation, with B the number of Monte Carlo permutations and J the number of predictors, which is negligible at the single-quarter horizon at which the decomposition is reported. The values ϕ_j satisfy $\hat{y} = \bar{y} + \sum_{j=1}^{J} \phi_j$ (Lundberg and Lee, 2017), providing an exact additive decomposition of the nowcast around the in-sample mean.

The user does not see this machinery: `outputNowcast` returns a ready-to-print `ggplot` waterfall chart (`shapley_plot`), the underlying contributions in tidy form (`norm_shapley`), and the baseline (`intercept`).

[5] The Monte Carlo evidence in Abdel-Latif et al. (2026) points the same way, with coverage above the nominal level and intervals wider than the oracle, that is, conservative intervals at the cost of width.

For analysts who want decompositions at multiple historical dates, useful for narrating turning points, a separate `shap_values` function offers per-date contributions.

2.7. Operational refresh cadence

Inside the IMF, nowcasts are not produced once: they are refreshed every time a relevant indicator is released, sometimes weekly within a quarter. Re-running the full hyperparameter search on every refresh is wasteful, because the predictor set has not changed, the topology of the loss surface has not moved much, and the analyst usually wants the new data point reflected in the existing model rather than a re-tuned one.

We introduce a `tuning = FALSE` mode for `macroNowcast`. The user passes the `all_results` object from a previous run as the `grid_results` argument; the function then skips the grid search, uses the previously selected hyperparameter configuration of each model, refits each finalized workflow on the latest data, and recomputes point forecasts, prediction intervals, SHAP values, and the model ranking on the same block-bootstrap CV plan. The full pipeline collapses to model fitting on a few dozen observations, more than an order of magnitude faster than the initial tuning pass (Section 7.2). This makes the package easy to embed in scheduled refresh scripts, and it is also the engine of the rolling-origin evaluation in Section 6.

2.8. The role of the analyst

It would be a misreading of the design above to conclude that nowcasting becomes a mechanical exercise. The package automates what should be automated: estimation, tuning, calibration, model comparison, and reporting, which are the steps where ad hoc practice is most error-prone and least reproducible. It does not choose the specification. The analyst assembles the predictor set from the country's release calendar and data infrastructure (which activity, trade, financial, survey, and satellite indicators exist, at what frequency, with what publication lag); decides the growth transformation of the target and the aggregation of higher-frequency indicators; inspects the automatic differencing decisions and overrides them where the diagnostics of Section 2.3 warrant it; chooses whether a late-released indicator should enter contemporaneously with imputation or lagged; and reads the ensemble, the MCS, the sensitivity table, and the SHAP decomposition before a number goes into a briefing. The Uganda predictor table in AppendixD and the sensitivity exercise of Section 4.5 are examples of that judgement being exercised and documented, not replaced.

3. Software architecture

Three guiding principles run through the package's design: i) *Reproducibility*: standardized functions replicate the IMF's NowHub methodology using publicly available tools, and every run is seeded; ii) *Flexibility*: a

curated but extensible set of model families can be customized for specific datasets and economic contexts; and iii) *Usability*: the interface abstracts away unnecessary complexity while remaining transparent for users who wish to inspect or extend the underlying code.

`IMFNow` exposes two top-level functions, whose shape follows directly from the design responses in Section 2:

- `macroNowcast(formula, data, ...)` runs the full pipeline (stationarity testing, block-bootstrap tuning, refitting, conformal interval construction, SHAP) and returns a `list` containing the fitted workflows, the best-model nowcast, the per-model nowcasts at all confidence levels, the out-of-sample calibration predictions, the differencing decisions, the SHAP decomposition, and the metadata needed to reproduce the run.
- `outputNowcast(nowcast_fit, country_iso3, ...)` consumes that object and returns the diagnostics needed to communicate the result: a prediction plot, a loss-comparison plot, a SHAP waterfall, the Model Confidence Set, and tidy data frames for custom plotting.

Two further utilities support the workflow. `train_test_split` partitions a panel by proportion, by number of holdout periods, or by date cutoff while respecting the temporal ordering. `extract_all_results` constructs averaged forecasts (weighted or unweighted) from arbitrary subsets of the fitted models, valuable for restricting the ensemble to the surviving members of an MCS.

Model suite. Internally, the suite is built on the `tidymodels` ecosystem (Kuhn and Wickham, 2020), with time-series-specific engines provided through `modeltime` (Dancho, 2023); among existing tidy-forecasting packages, the closest analogue on the user-facing side is `fable` (O'Hara-Wild et al., 2024), which exposes a unified interface to a curated set of classical time-series models but does not target the short-sample, model-selection-uncertainty regime that motivates `IMFNow`. The default model list (`.default_models`, nine models) covers Elastic Net regression (`Elast`), Partial Least Squares (`PLS`), a random forest with block-bootstrap resampling of the time index (`RF`), XGBoost (`XGboost`), Bayesian Additive Regression Trees (`BART`), Cubist rule-based regression (`Cubist`), Neural Network Autoregression (`NNETAR`), an ARIMA benchmark with the predictors as exogenous regressors (`AR`, an ARIMAX whose orders are selected by BIC), and a Dynamic Factor Model (`DFM`). An extended list (`.extended_models`, sixteen models) adds three SVM variants, *k*-nearest neighbors, decision trees, bagged trees, and a multilayer perceptron (MLP). Both lists are ordinary R `list` objects: users can subset them, extend them with custom `tidymodels` specifications, or replace them entirely. The ML models share a normalized `recipe` (imputation, then centring and scaling of the predictors) with `date` marked as an ID variable; the time-series models retain `date` as a predictor as required by their `modeltime` engines.

The DFM engine. Because the dynamic factor model is the workhorse of the nowcasting literature, its treatment of the ragged edge deserves a precise statement. The engine wraps `dfms` (Krantz and Bagdziunas, 2024): the target and the (imputed, normalized) predictors are stacked into one panel, r factors with a VAR(p) transition are estimated by the EM algorithm with Kalman smoothing (Doz et al., 2011), and r and p are tuned like any other hyperparameter. For a nowcast row, the fitted model's Kalman filter and smoother are run over the training panel extended by the new row with the target set to missing, with all system matrices held fixed, and the prediction is the smoothed common component of the target in that row. The DFM nowcast therefore conditions on the same-quarter predictor values through the measurement update, as a factor-model nowcast should; no parameter is re-estimated at prediction time. Because the predictors reach the DFM after the recipe's imputation, the model's native ability to handle missing predictor values through the filter is not used; we keep one imputation path for all engines for simplicity and comparability.[6]

Lags and hyperparameters. Lags of the target and of the predictors enter as explicit columns controlled by the user through `lags_y` and `lags_x` (default 0); they are constructed once, before the stationarity step, and are visible to every engine. Three engines add dynamics of their own: the ARIMAX selects its own autoregressive and moving-average orders by BIC, NNETAR tunes the number of autoregressive inputs (fixed at zero when `lags_y > 0`, to avoid duplicating the user's lags), and the DFM tunes the order of its factor VAR. When `lags_y > 0` and more than one trailing row is missing, the nowcasts are produced recursively, each missing quarter's prediction feeding the lag of the next. Hyperparameters are selected by cross-validation: for each model, `grid_levels` candidate configurations are drawn from a space-filling design over the tunable parameters (`tune::tune_grid` with an integer grid), every candidate is evaluated on every block-bootstrap resample, and the configuration with the best mean `metric_select` is refitted on the full sample. The tunable parameters and their ranges are listed in AppendixC.

Frequencies and sample sizes. The pipeline is frequency-agnostic: it sees a `Date` column and rows, so monthly targets (a monthly GDP indicator, for example) work without changes, and higher-frequency predictors enter in one of two ways, at the user's choice: aggregated to the target's frequency before estimation, so that mixed frequencies collapse into the ragged-edge problem and are handled by the trailing-NA convention, or left disaggregated with their individual high-frequency lags entered as separate regressors with unconstrained coefficients, the unrestricted MIDAS (U-MIDAS) construction of Foroni et al. (2015), which dispenses with the lag polynomials of standard MIDAS regressions. Such terms are additional columns in the design matrix and pass to the engines unchanged; the paper does not present a worked U-MIDAS example, and an automated layer that would build the lag terms from raw data is left to future work. Annual targets are possible in principle but not advisable at the sample sizes available in practice: with 25–30 annual observations the

[6] Versions before 0.0.6.0 returned an h-step factor forecast for new rows that ignored their predictor values, so the DFM column of earlier horse races was a forecast rather than a nowcast.

block length is capped at six or seven, the first calibration window leaves fewer than 20 conformity scores (so the 95% and 99% intervals are infinite), and most of the learners in the suite cannot be tuned meaningfully. Quarterly samples of 40 observations or more, or monthly samples of 100 or more, are the regime the package is designed for.

3.1. Installation

The latest version of `IMFNow` can be installed from its GitHub repository:

```
require(devtools)
devtools::install_github("hanomicsIMF/IMFNow", dependencies = TRUE)
```

The package is then loaded into the workspace with

```
library(IMFNow)
```

Three vignettes (*getting started*, *models and setup*, *interpreting results*) play the role of a how-to note and are the recommended entry point for new users.

4. GDP nowcast for Uganda

We illustrate the workflow on a quarterly GDP nowcast for Uganda, using each step to point back to the design responses in Section 2, demonstrating that the design works end-to-end on a realistic dataset.

4.1. Data

The target is Uganda's real GDP growth in year-on-year terms to account for seasonality without a seasonal-adjustment step whose parameters would themselves be unstable in a short sample. The sample runs from 2013Q1 to 2026Q2, with GDP observed for 53 quarters and missing in the last one, which is the quarter to be nowcast. The dataset is taken from the IMF NowHub data pipeline (vintage 2026Q2, built on 12 August 2026), which collects monthly indicators from various sources including Haver Analytics, IMF databases, national authorities, and satellite and web sources. The data pipeline then aggregates these monthly indicators to calendar quarters, and applies a documented transformation to each series (year-on-year changes for flows, stocks and most indices; levels for survey balances, anomalies and the uncertainty index). The 14 predictors were chosen from the indicators that exist for the country with a usable publication lag: five series disseminated by the Bank of Uganda (a composite index of economic activity,

Table 1: Last five quarters of the Uganda panel passed to `macroNowcast`

Variable	2025Q2	2025Q3	2025Q4	2026Q1	2026Q2
Real GDP growth, y/y (target)	5.53	4.85	8.46	5.79	NA
Composite index of economic activity	0.07	0.07	0.07	0.07	0.05
Business indicator	59.27	58.10	56.73	57.53	55.67
Domestic credit	0.12	0.05	0.04	0.08	0.07
Exports	0.55	0.41	0.87	0.56	0.17
Real effective exchange rate	0.05	0.06	0.02	-0.02	-0.03
Terms of trade	0.28	-0.15	0.01	-0.05	-0.23
PMI, China	-0.01	0.00	-0.01	-0.01	0.02
World Uncertainty Index	0.05	0.12	0.08	0.16	0.10
Google Trends, tourism	12.67	40.67	70.67	65.33	65.00
Aerosol optical depth	-0.15	0.12	-0.18	-0.48	-0.03
NDVI anomaly	0.70	0.66	0.66	0.59	0.69
NDWI	-0.91	-0.85	-1.14	-1.04	-1.03
NO2 column density	-0.03	0.37	0.18	-0.19	0.26
Night-time lights	-0.01	1.44	-0.14	-0.37	-0.17

Notes: all 15 columns of the data frame, transposed. The target is missing in 2026Q2, the quarter to be nowcast; every predictor is observed for that quarter. Predictors are as supplied by the IMF NowHub data vintage of 2026Q2 (year-on-year changes for flows, stocks and most indices; levels for the business indicator, the uncertainty index, the tourism index and the vegetation and water indices). AppendixD gives the source and transformation of each series and the differencing decision taken by the pipeline.

a business tendency indicator, domestic credit, exports, and the terms of trade), the IMF's CPI-based real effective exchange rate, China's manufacturing PMI as the external-demand proxy, the World Uncertainty Index, a Google Trends index for tourism, and five satellite-derived series (MODIS aerosol optical depth, a vegetation-index anomaly, a surface-water index, tropospheric nitrogen-dioxide density from TROPOMI and OMI, and VIIRS night-time lights). AppendixD lists each predictor with its source, native frequency, the transformation applied in the vintage, and the pipeline's differencing decision; the KPSS step differenced 4 of them (exports, the exchange rate, the tourism index, and the water index) and left the target in levels. Table 1 shows the last five quarters of the panel: the most recent quarter has `gdp = NA` and every predictor observed, the trailing-NA convention of Section 2.

No further metadata is required: `macroNowcast` will detect the missing target row, hold it aside as the prediction set, and return one nowcast per missing trailing period.[7]

4.2. A minimal call to `macroNowcast`

`macroNowcast` is the main estimation function in `IMFNow`, designed to produce nowcasts by horse-racing a user-specified set of candidate models. Given a target variable, a set of predictors, and a list of model specifications, it estimates, tunes, compares, and refits the models before returning point nowcasts, prediction intervals, and model diagnostics. The call that produced every Uganda result in this paper is

[7] If more than one observation is missing at the end of the sample, for example because both the latest and the previous release have been delayed, the function returns nowcasts for all such trailing missing periods.

```
fit <- macroNowcast(gdp ~ ., data = uga_data,
                    metric_select = "rmse", conf_level = 0.90,
                    cv_param = list(reps = 49, block_size = NULL),
                    grid_levels = 25, seed = 1234)
out <- outputNowcast(fit, country_iso3 = "UGA")
```

`macroNowcast` requires only a `formula` and a `data` argument. We use `gdp ~ .`, with `gdp` as the dependent variable and `.` taking all remaining columns as regressors; a subset of predictors can be specified explicitly, e.g. `gdp ~ reer + trade_terms + wui`. Although the example focuses on GDP, the function is agnostic to the target variable.

The remaining arguments map onto the design responses described above. The model suite is controlled by `models_list`, the list of `tidymodels` specifications to horse-race; here we use the default nine, but subsetting via e.g. `.default_models[c("Elast","RF","DFM")]` or replacing with `.extended_models` is straightforward. Lag structure is controlled by `lags_y` and `lags_x` (default 0, used here). Model selection is driven by `metric_select`, the metric used to tune, rank and select the models on the block-bootstrap resamples; it is one of seven metrics computed on every resample: RMSE (the default, used throughout this paper so that selection, the loss plot, the ensemble weights and the MCS share one criterion), MAE, Huber and pseudo-Huber loss, R^2, the concordance correlation coefficient (CCC, which rewards correct identification of turning points), and the package's own `macf`, the absolute first-order autocorrelation of the forecast residuals. Cross-validation is controlled by `cv_param`, which configures the block-bootstrap CV plan (49 replications and a data-selected block length here), and by `grid_levels`, which sets the number of candidate configurations per model (25 here). The stationarity treatment is controlled by `difference` (default `"auto"`) and the treatment of missing predictors by `na_handling` (default `"impute_linear"`). Reproducibility is enforced by `seed`.

Interval construction is governed by `conf_level`, the level reported in the best-model summary (90% here; the four standard levels remain available in `nowcast_all_models`), `conf_method` (default `"jackknife"`, the expanding-window calibration; `"split"` for the faster split-conformal alternative), `individual_multiplicative` (default `FALSE`: per-model intervals are additive, the ensemble interval multiplicative; `TRUE` applies the studentised calibration to every model), and `top_k` (default 10), the number of hyperparameter configurations per model that form the pseudo-population of the multiplicative calibration. The operational fast-update mode discussed in Section 7.2 is enabled by `tuning` and `grid_results`. Finally, `compute_intervals` and `compute_shap` (both default `TRUE`) can be set to `FALSE` to skip conformal calibration or SHAP decomposition when only point forecasts are needed, which is useful for large-scale model-selection studies. Tuning and calibration windows run in parallel under a user-supplied `future` plan (Bengtsson, 2021).

The function returns a `list` whose principal elements are the best-model point and interval nowcasts (`nowcast`), per-model intervals at every confidence level (`nowcast_all_models`), the out-of-sample predictions of every model at the calibration dates (`nowcast_oos`), the name and tuning parameters of the best model (`best_model_name`, `best_model_tuning_params`), the final fitted workflow (`final_fit`), SHAP values for the best model (`shap`), the differencing decisions (`differencing`), the full tuned workflow set (`all_results`, reusable via `grid_results`), and the formula and data used. See `?macroNowcast` for the complete list.

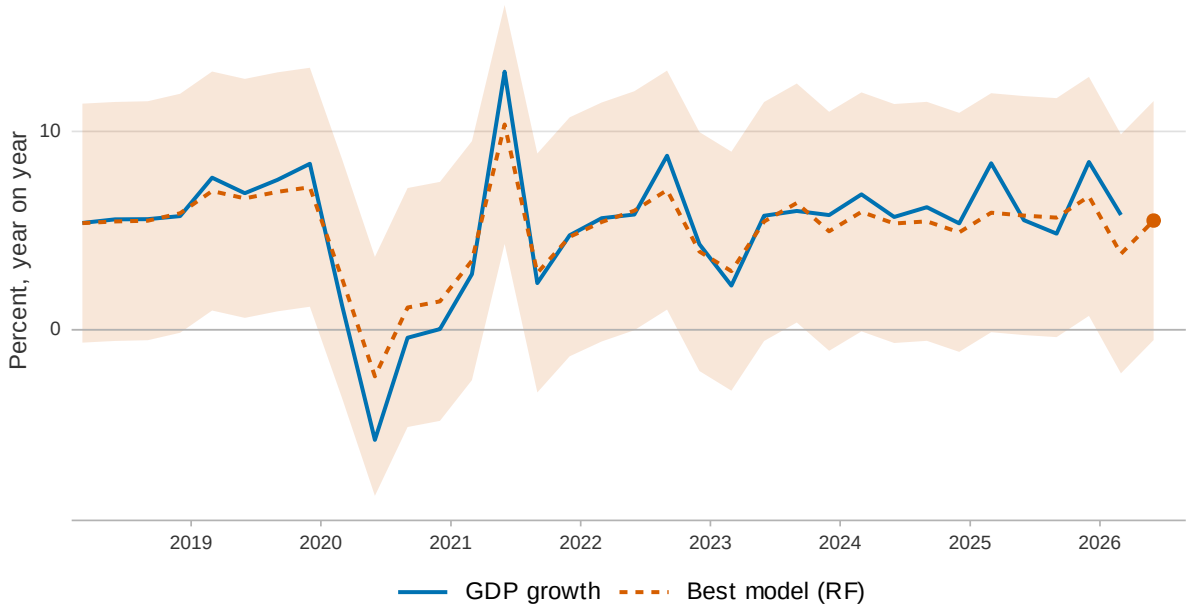
4.3. Nowcast diagnostics

Once `macroNowcast` has produced the model horse-race, `outputNowcast` consumes the fitted object and returns the predictions, plots, and tests required to assess the results. The function takes as main arguments the output of `macroNowcast` (`nowcast_fit`) and the country ISO3 code (`country_iso3`). `start_date` (default `2018-01-01`) truncates the displayed series; `save_path` sets the output directory; `export_csv` writes the main results to CSV; `display_plots` prints plots during execution; and `save_plots` writes `.png` files. The return value is a list whose contents directly mirror the model-selection-uncertainty machinery of Section 2:

- `nowcast_results`: actuals and best-model predictions with the prediction interval at the user-specified `conf_level`, filtered to dates from `start_date` onward. This is the primary object for a single-model summary.
- `all_models_nowcast`: the same information for *every* fitted model and the ensemble, at all four standard confidence levels; a `conf_level` column identifies the nominal level. This is the object that powers the fan chart below.
- `prediction_plot` (Figure 1): the best model's nowcast against actuals, with its prediction interval.
- `loss_plot` (Figure 3): the cross-validated metric of every model, with its standard error across the block-bootstrap replications.
- `MCS_results`, `MCS_survivors`, `MCS_loss_basis`, `MCS_n_obs`: the Model Confidence Set, the models it contains, the loss basis (out of sample, or the in-sample fallback) and the number of dates it was computed on.
- `norm_shapley` and `shapley_plot`: the SHAP decomposition of the last missing quarter, in tidy form and as a waterfall plot. `intercept` is the in-sample mean.

The best model by cross-validated RMSE is the block-bootstrap random forest (RF), with a CV RMSE of 2.15 percentage points (standard error 0.21 across the 49 replications). Its nowcast of year-on-year GDP growth for 2026Q2 is 5.51%, with a 90% prediction interval of $[-0.52, 11.53]$. Figure 1 plots this nowcast at the end of the series. The historical line of the best model in Figure 1 is the *in-sample* fit of the model refitted on the whole

Figure 1: Uganda: year-on-year GDP growth and the best-model nowcast



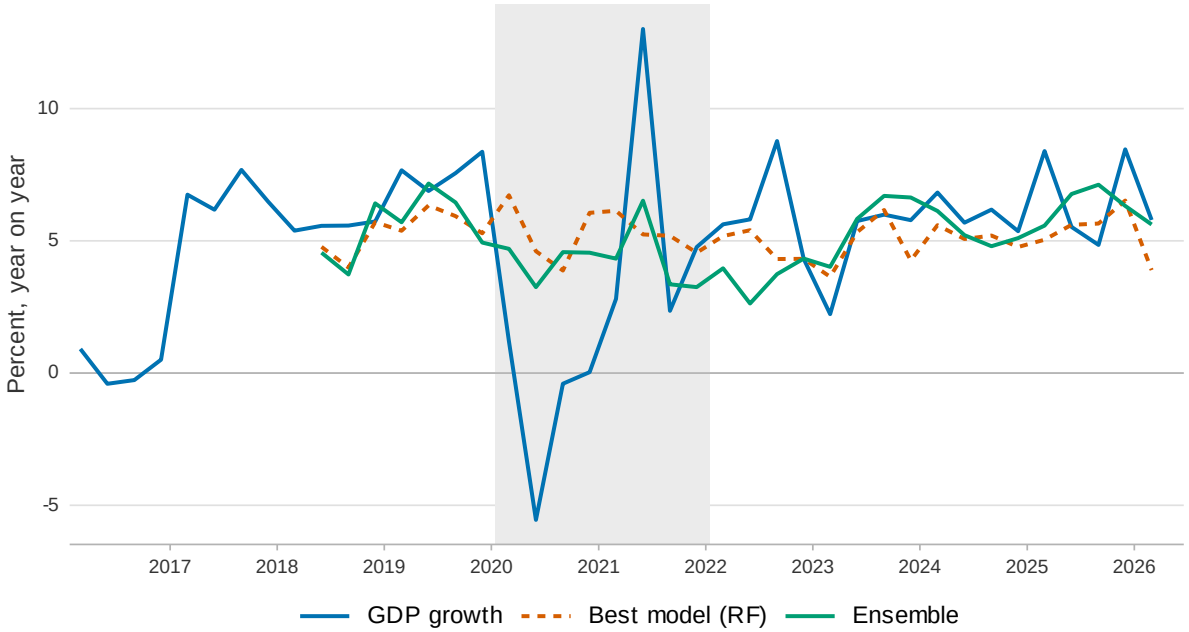
Notes: Actual year-on-year real GDP growth (solid) and the nowcast of the best model by cross-validated RMSE (RF, dashed), 2018Q1 to 2026Q2. The shaded band is the 90% expanding-window conformal prediction interval, drawn with the same critical value around the in-sample fit for past quarters and around the nowcast at 2026Q2. For quarters before 2026Q2 the dashed line is the in-sample fit of the model estimated on the full sample; the marker at 2026Q2 is the nowcast (5.51%, interval [−0.52, 11.53]). Out-of-sample predictions are shown in Figure 2. Source: authors' calculations with `IMFNow` 0.0.7.1.

sample, which is why it tracks the actuals so closely. The out-of-sample record is in Figure 2, which plots, for the same best model and for the ensemble, the prediction made for each quarter from 2016Q1 to 2026Q1 by the model refitted on data strictly before that quarter (the expanding-window calibration path of Section 2). The out-of-sample errors are larger, as they should be: the best model's RMSE over the 32 calibration quarters at which it produced a prediction is 3.24 percentage points against 2.15 in cross-validation. The pandemic quarters account for most of the gap: excluding 2020Q1 to 2021Q4 (shaded in the figure), the out-of-sample RMSE is 1.68 for the best model and 1.90 for the ensemble, close to the cross-validated figures, which is the comparison a desk should make when judging the models' accuracy in ordinary quarters. The full comparison of cross-validated and out-of-sample RMSE by model is in AppendixE.

Figure 3 shows that seven of the nine models have a cross-validated RMSE between 2.1 and 2.5 percentage points, differences of the same order as the standard errors (about 0.2), and only the neural-network autoregression and the ARIMAX benchmark are clearly worse. We therefore also report the ensemble nowcast (Figure 4) with its 90% prediction interval. The best-model and ensemble point forecasts are close, 5.51% and 5.21%, and the ensemble interval, [−3.32, 10.38], is of similar width to the best model's but shifted down, reflecting the models' disagreement on the downside in the calibration history.

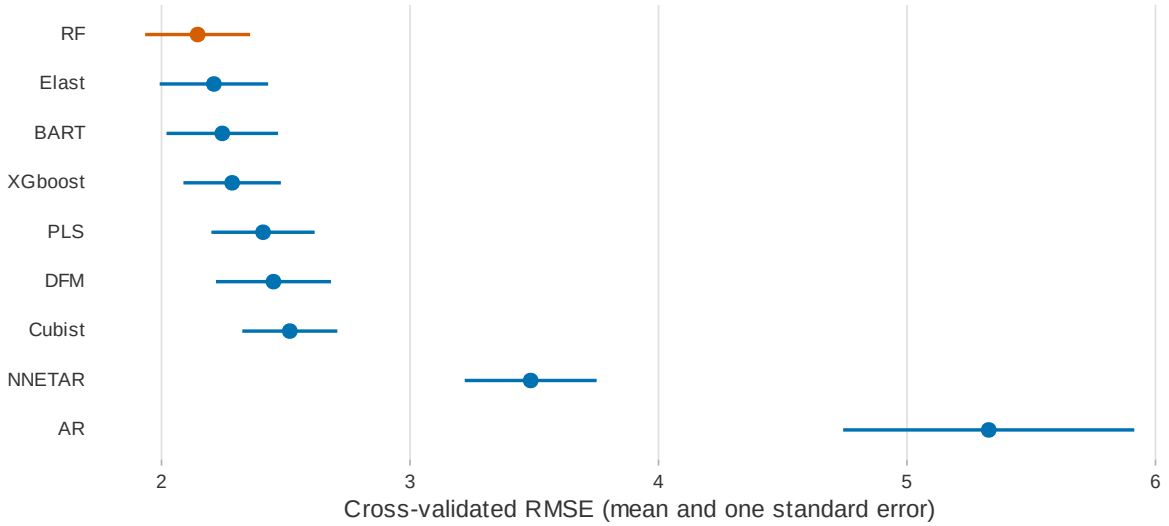
The Model Confidence Set in Table 2 formalizes the same intuition. It is computed at the 90% level on

Figure 2: Uganda: expanding-window out-of-sample predictions



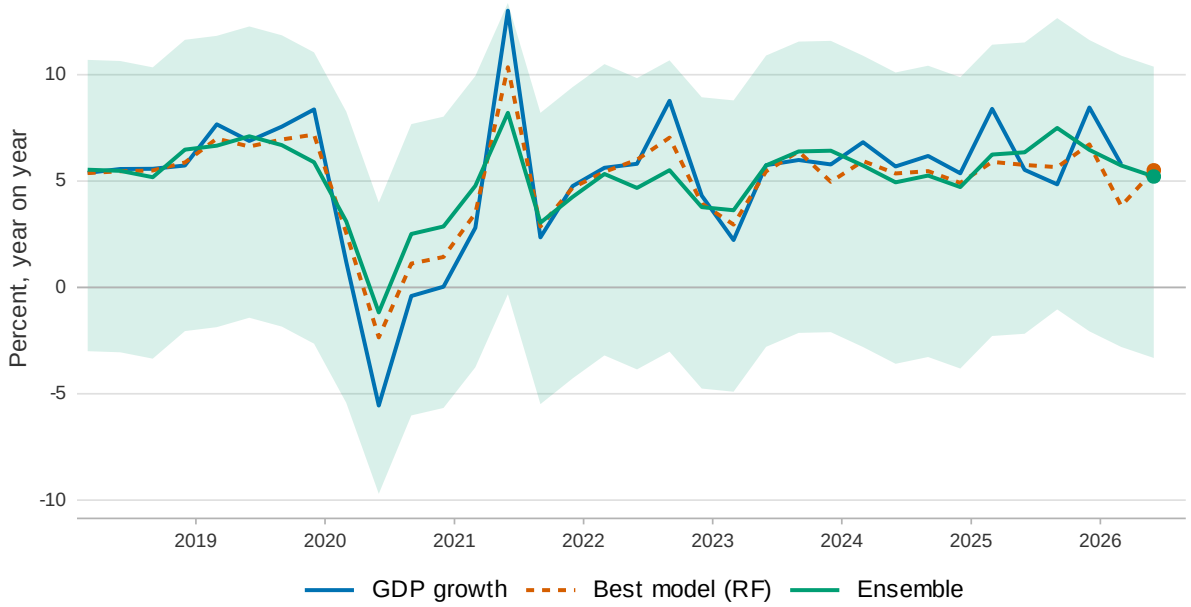
Notes: For each quarter from 2016Q1 to 2026Q1, the prediction of the best model (RF, dashed) and of the ensemble (solid) made by the models refitted on data strictly before that quarter, against actual growth. These are the predictions that calibrate the conformal intervals and feed the Model Confidence Set. The shaded band marks the pandemic quarters 2020Q1 to 2021Q4, which carry the largest errors and set the width of the 95% intervals; the out-of-sample RMSE with and without them is in AppendixE. The ensemble line starts in 2018Q2, the first calibration date at which every model produced a prediction (the random forest's selected block length exceeds the shortest windows). Source: authors' calculations.

Figure 3: Uganda: cross-validated RMSE of the candidate models



Notes: Mean RMSE across the 49 block-bootstrap replications, with one standard error on either side, for the nine candidate models; the best model (RF) is highlighted. Lower is better. Source: authors' calculations.

Figure 4: Uganda: best-model and ensemble nowcasts



Notes: Actual year-on-year real GDP growth, the best model's nowcast (RF, dashed) and the inverse-RMSE-weighted ensemble of the nine models (solid), 2018Q1 to 2026Q2. The shaded band is the ensemble's 90% prediction interval with multiplicative PCS-UQ calibration, drawn around the in-sample ensemble for past quarters. Historical lines are in-sample fits; the markers at 2026Q2 are the nowcasts (5.51% and 5.21%). Source: authors' calculations.

the squared out-of-sample errors of the nine models and the ensemble over the 32 calibration quarters at which every candidate produced a prediction. The table should be read as follows. The first column is each candidate's average squared error; the second is the p-value of the equal-predictive-ability test at the step at which the candidate was, or would have been, eliminated; the third is the MCS p-value, the running maximum of the second column; a candidate is in the 90% MCS when its MCS p-value is at least 0.10. In this example exactly one model is eliminated, the ARIMAX benchmark (MCS p-value 0.05), whose average squared error is four to five times that of the others; the remaining 9 candidates, the ensemble included, cannot be distinguished at this sample size. That is the typical outcome in a panel of fifty quarters, and it is the formal case for averaging rather than selecting: the function `extract_all_results` accepts a `model_names` argument that restricts the ensemble to a subset (for example the survivors of the MCS), a `weights` argument that toggles between inverse-loss weighting (the default) and a simple average, and a `conf_levels` argument controlling the confidence levels at which prediction intervals are computed.

4.4. Variable importance via SHAP values

The point of the SHAP machinery is communication: a desk economist who has to defend a country briefing needs to be able to attribute the nowcast to specific economic drivers, even when the underlying model is a random forest. Figure 5 shows the SHAP decomposition of the best model's nowcast for 2026Q2. The

Table 2: Uganda: 90% Model Confidence Set on out-of-sample squared errors

Model	Avg. squared error	p-value, H_{0,M_k}	MCS p-value	In the 90% MCS
AR	41.92	0.05	0.05	no
DFM	13.53	0.51	0.51	yes
BART	10.63	0.60	0.62	yes
Cubist	11.94	0.33	0.62	yes
NNETAR	13.16	0.62	0.62	yes
RF	10.50	0.66	0.66	yes
Elast	8.54	0.95	0.98	yes
PLS	8.90	0.98	0.98	yes
Ensemble	8.41	0.94	0.98	yes
XGboost	8.33	1.00	1.00	yes

Notes: Procedure of Hansen et al. (2011) as implemented in the `MCS` package (5,000 block-bootstrap replications, seeded), applied to the squared errors of each candidate's expanding-window out-of-sample prediction over the 32 quarters at which all ten candidates produced a prediction (2018Q2 to 2026Q1). Models are listed in order of elimination. "p-value, H_{0,M_k}" is the equal-predictive-ability test p-value at the step in which model M_k was or would have been eliminated; the MCS p-value is the running maximum of that column; a model is in the 90% MCS when its MCS p-value is at least 0.10. Source: authors' calculations.

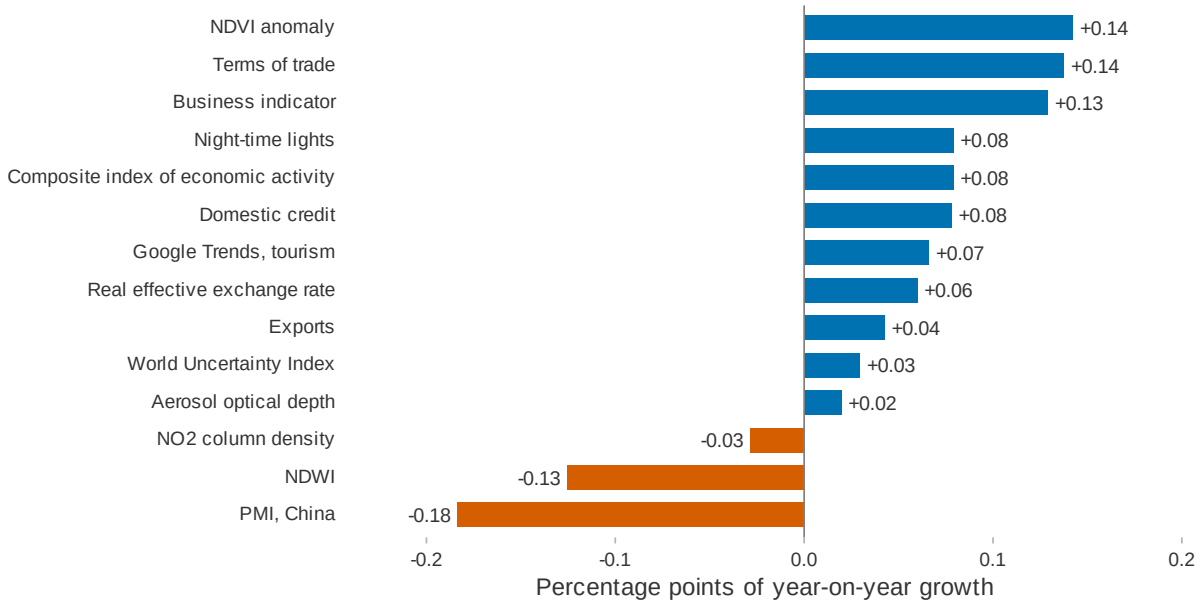
horizontal axis measures each variable's contribution in percentage points of year-on-year growth; the baseline is the in-sample mean of 4.98%; and each bar reports how much a given predictor's observed 2026Q2 value shifts the nowcast away from the baseline, the contributions summing exactly to the nowcast of 5.51%.

In this quarter the contributions are individually small, which is itself informative: the nowcast is close to the historical mean because no indicator departs strongly from its typical value. The largest negative contribution comes from the weaker Chinese PMI (external demand), partly offset by the vegetation anomaly, the terms of trade, and the business indicator. This decomposition is what makes the best model usable as a black box: analysts can attribute the nowcast to specific economic drivers without inspecting the forest's internal parameters. The normalized contributions are returned as the `norm_shapley` data frame for further analysis or custom plotting; for narratives that span several historical dates, a separate `shap_values` function provides per-date Shapley analysis.

4.5. Sensitivity to a late predictor release

The previous subsections assumed that every predictor in the nowcast quarter had been observed. In an operational setting this is rarely the case, and Section 2.2 described how the pipeline proceeds when a predictor is missing: by default it imputes the missing value inside the fitted workflow's recipe. The natural question for a desk is how much the nowcast depends on that treatment, predictor by predictor, because the answer tells the analyst which releases are worth waiting for and which can be imputed without consequence. Table 3 answers it for Uganda without re-tuning or refitting anything: for each of the 14 predictors in turn, its 2026Q2 value is set to `NA` and the already-fitted best-model workflow is asked to predict, once with the recipe's linear imputation (the `"impute_linear"` default, which regresses the missing predictor on the other thirteen) and once with the previous quarter's value carried forward (`"forward_fill"`; a zero first difference

Figure 5: Uganda: SHAP decomposition of the best-model nowcast for 2026Q2



Notes: Sampling Shapley values (Štrumbelj and Kononenko, 2014) of the RF nowcast for 2026Q2, 100 Monte Carlo permutations, seeded. Each bar is the contribution of one predictor, in percentage points, to the difference between the nowcast (5.51%) and the in-sample mean (4.98%). Positive (negative) bars push the nowcast above (below) the historical mean. Source: authors' calculations.

for the series the pipeline differenced). The table reports the shift of the nowcast relative to its value on the fully observed row (5.51%).

The shifts are small. Under linear imputation the largest is −0.32 percentage points (the business indicator), and no predictor moves the nowcast by more than a third of a point; the World Uncertainty Index, the example of a late release used in the earlier version of this paper, moves it by 0.06 under imputation and by −0.69 under forward fill. Two lessons follow. First, the shift measures the distance between the imputed and the realised value of a predictor times the model's response to that predictor, so it is a diagnostic of imputation risk, not a measure of the predictor's information content; a predictor can matter greatly for the nowcast and still show a small shift if the other predictors forecast it well. Second, the treatment matters more for some series than the fact of missingness: forward-filling night-time lights or the uncertainty index moves the nowcast by more than half a point because both series are volatile quarter to quarter, whereas imputing them from the other predictors does not. The exercise therefore supports the default of imputation for occasional gaps, and the lagged-predictor construction of Section 2.2 for systematically late series. It is cheap enough (fourteen predictions from a fitted workflow) to be repeated at every refresh, and the analyst can extend it to combinations of predictors or to the ensemble.

Table 3: Uganda: shift of the best-model nowcast when one predictor is missing in 2026Q2 (percentage points)

Predictor missing in the nowcast quarter	Linear imputation	Forward fill
Business indicator	-0.32	0.05
NDVI anomaly	-0.27	-0.23
Exports	-0.19	-0.13
Google Trends, tourism	-0.11	-0.02
Night-time lights	0.10	-0.83
Aerosol optical depth	0.08	0.16
NO2 column density	0.07	0.09
Terms of trade	-0.07	0.00
PMI, China	0.06	0.23
World Uncertainty Index	0.06	-0.69
Composite index of economic activity	-0.03	0.13
Domestic credit	0.02	-0.00
NDWI	0.00	-0.08
Real effective exchange rate	0.00	0.00

Notes: For each row the named predictor's 2026Q2 value is set to missing and the fitted RF workflow (Section 4.3) predicts the quarter again, with the missing value either imputed by the recipe's linear regression on the other predictors (`na_handling = "impute_linear"`) or replaced by its 2026Q1 value (`"forward_fill"`; zero for differenced series). Entries are the resulting nowcast minus the nowcast on the fully observed row (5.51%). No model is re-tuned or refitted. Rows sorted by the absolute imputation shift. Source: authors' calculations.

4.6. Forecasting beyond the last sample period

The trailing-NA convention introduced in Section 2 extends straightforwardly to forecasts beyond the last observed quarter. To produce an h-step-ahead forecast given information available at time t, we apply a direct multi-step strategy: lead the target by h quarters (so that the regression at row t relates regressors at $t - h$ to GDP at t) and append h extra rows at future dates that hold the most recent regressors and `gdp = NA`. With $h = 1$ in the Uganda example, the two trailing rows of the data have `gdp = NA` (Table 4): one for the quarter that motivated the original nowcast, one for the quarter beyond it.

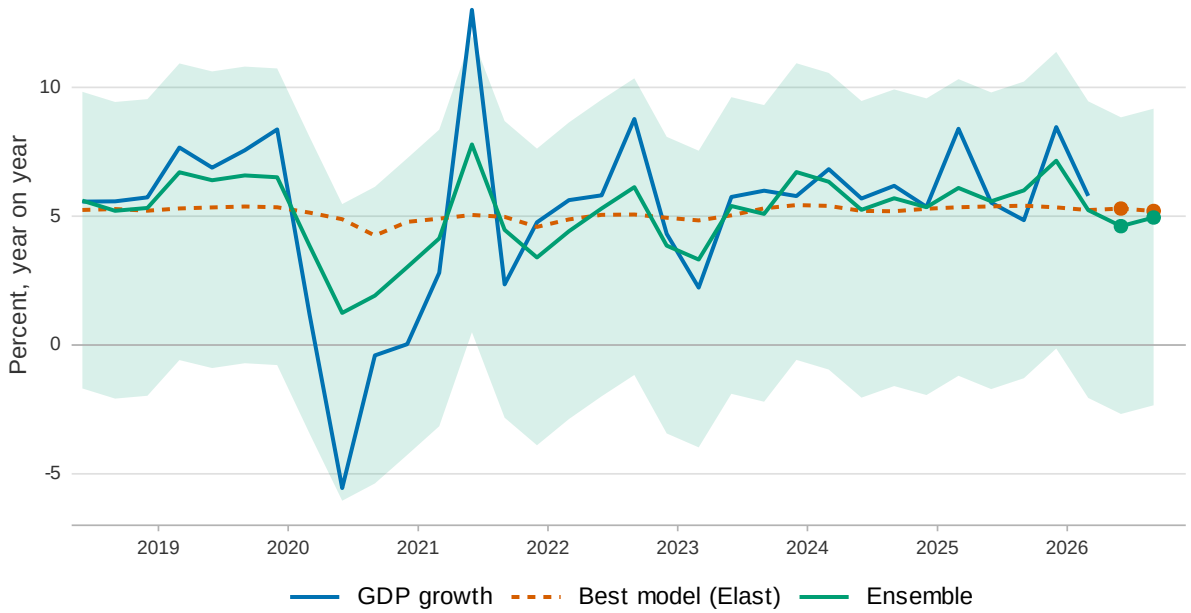
The rest of the workflow is unchanged: the same call to `macroNowcast` followed by `outputNowcast` on the modified data produces the forecasts for both missing periods, with the tuning repeated on the lead-transformed panel. The best model on this panel is the Elastic Net, whose forecast for 2026Q3 is 5.20%, against 4.95% for the ensemble (Figure 6). The figure also shows what the selection means on this panel. The cross-validated losses of the leading candidates are within one standard error of each other (2.30 for the Elastic Net against 2.31 and 2.32 for the DFM and BART, with standard errors of about 0.24), the Model Confidence Set retains all 10 candidates, and the Elastic Net was selected at the most penalised point of its tuning grid: its fitted values lie between 4.2% and 5.4% in every quarter shown, while actual growth ranged from −5.6% to 13.0%, so its forecast is close to the sample mean of Ugandan growth. One quarter ahead, these indicators add little to that mean, and the ensemble, which mixes in the factor model and the tree-based learners, is the more informative line; the flat dashed line is the diagnostic, not a defect. The remaining diagnostics (loss comparison, MCS, SHAP) are produced exactly as in the one-step case.

Table 4: Last five quarters of the Uganda panel after the one-quarter lead transformation

Variable	2025Q3	2025Q4	2026Q1	2026Q2	2026Q3
Real GDP growth, y/y (target)	4.85	8.46	5.79	NA	NA
Composite index of economic activity	0.07	0.07	0.07	0.07	0.05
Business indicator	59.27	58.10	56.73	57.53	55.67
Domestic credit	0.12	0.05	0.04	0.08	0.07
Exports	0.55	0.41	0.87	0.56	0.17
Real effective exchange rate	0.05	0.06	0.02	-0.02	-0.03
Terms of trade	0.28	-0.15	0.01	-0.05	-0.23
PMI, China	-0.01	0.00	-0.01	-0.01	0.02
World Uncertainty Index	0.05	0.12	0.08	0.16	0.10
Google Trends, tourism	12.67	40.67	70.67	65.33	65.00
Aerosol optical depth	-0.15	0.12	-0.18	-0.48	-0.03
NDVI anomaly	0.70	0.66	0.66	0.59	0.69
NDWI	-0.91	-0.85	-1.14	-1.04	-1.03
NO2 column density	-0.03	0.37	0.18	-0.19	0.26
Night-time lights	-0.01	1.44	-0.14	-0.37	-0.17

Notes: The row dated t holds GDP growth of quarter t and the predictors of quarter $t-1$. The target is missing for 2026Q2 (not yet released) and for 2026Q3 (one quarter ahead); both rows are nowcast by the same call under the direct multi-step strategy with $h = 1$.

Figure 6: Uganda: nowcast and one-quarter-ahead forecast



Notes: Actual year-on-year real GDP growth, the best model on the lead-transformed panel (Elast, dashed) and the ensemble (solid), with the ensemble's 90% prediction interval. The last two markers are the predictions for 2026Q2 and 2026Q3, made with regressors observed up to 2026Q2. Historical lines are in-sample fits. Source: authors' calculations.

Table 5: Tunisia and Uganda: headline outputs of the same call

	Tunisia	Uganda
Observed quarters (2013Q1 to 2026Q1)	53	53
Predictors (of which differenced)	14 (4)	14 (4)
Best model by CV RMSE	PLS	RF
CV RMSE of the best model (s.e.)	2.43 (0.33)	2.15 (0.21)
Best-model nowcast, 2026Q2 (%)	2.64	5.51
90% interval	[−1.16, 6.44]	[−0.52, 11.53]
Ensemble nowcast, 2026Q2 (%)	2.38	5.21
90% interval	[−16.56, 12.50]	[−3.32, 10.38]
Models eliminated from the 90% MCS	none	AR
Full tuning pass (minutes)	116.6	119.9

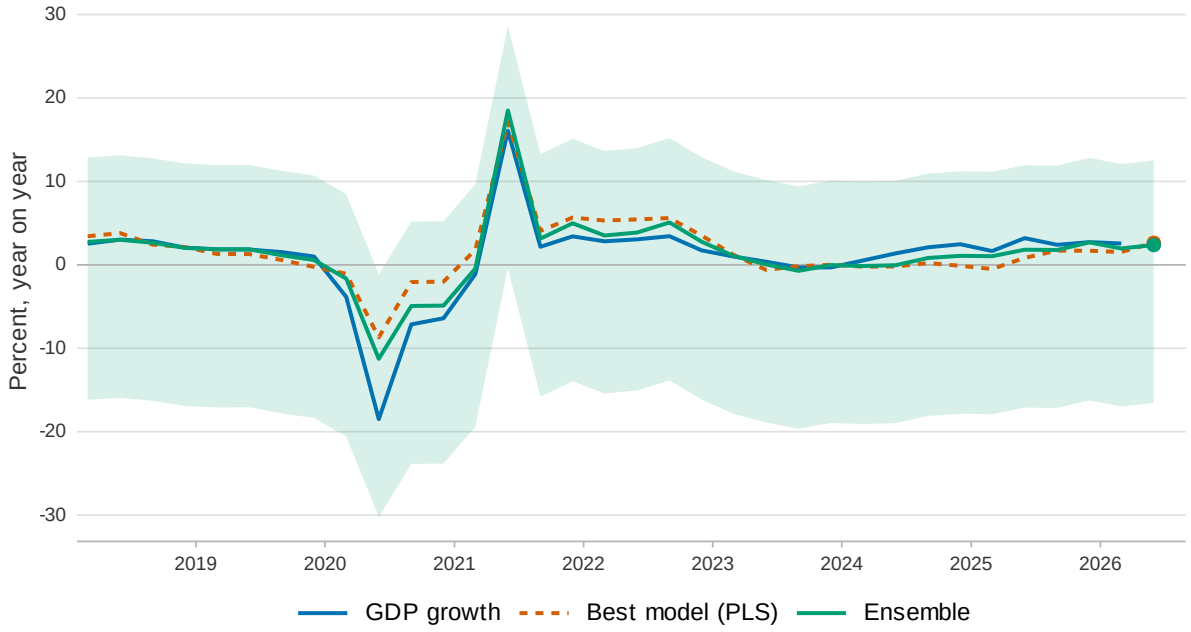
Notes: Identical `macroNowcast` settings for both countries (RMSE selection, 49 block-bootstrap replications, 25 grid points per model, seed 1234, automatic differencing, linear imputation). Timings on the same host (Section 7.2). The Tunisia MCS table and predictor list are in AppendixD and AppendixE. Source: authors' calculations.

5. A second application: Tunisia

To show the workflow in a different data environment we ran the identical call (same settings, same seed) on a Tunisia panel of the same length, with 14 predictors from the same data vintage: five national series via Haver Analytics (imports, exports, broad money, industrial production, and arrivals of travellers), the real effective exchange rate, the oil price, the World Uncertainty Index, a Google Trends index for tourism, and the same five satellite series (AppendixD; the KPSS step differenced 4 series). The headline outputs are in Table 5 and Figure 7.

Two features of the Tunisian result illustrate what the diagnostics are for. First, the best model is a linear one (partial least squares) and no candidate is eliminated from the MCS: the ten candidates, the ensemble included, are indistinguishable on 36 out-of-sample quarters. Second, the ensemble's 90% interval, [−16.56, 12.50], is far wider than the best model's, [−1.16, 6.44]. The reason is visible in the out-of-sample record (AppendixE): the two regularised linear models, which cross-validate well, produced extreme predictions in the pandemic quarters when they were refitted on the pre-pandemic history (the Elastic Net predicted growth above 100% for 2021Q2, the quarter of the base effect after the 2020Q2 collapse of −18.5%), and the multiplicative calibration of the ensemble interval, which scales with the dispersion of the candidate predictions, inherits those episodes through its conformity scores. The best model's additive interval is calibrated only on its own errors and is much narrower. Outside the pandemic quarters the linear models are not the problem: excluding 2020Q1 to 2021Q4, the out-of-sample RMSE is 1.46 percentage points for the best model and 0.90 for the ensemble, against 10.30 and 4.47 over the full calibration window (AppendixE). The two intervals are therefore answering different questions, the second being how wrong the whole suite has been when its members disagreed, and the honest reading of the Tunisia case is that the suite's disagreement in the pandemic quarters still dominates the ensemble's calibration five years later. Section 7.1 returns to this point; the practical response is in the analyst's hands (a shorter calibration sample, exclusion of the pandemic

Figure 7: Tunisia: best-model and ensemble nowcasts



Notes: As Figure 4 for Tunisia: actual year-on-year real GDP growth, the best model (PLS, dashed) and the ensemble (solid), with the ensemble's 90% prediction interval; markers at 2026Q2 are the nowcasts (2.64% and 2.38%). Historical lines are in-sample fits. Source: authors' calculations.

quarters, or the MCS-restricted ensemble), and the package makes the diagnosis rather than hiding it.

6. Pseudo-real-time evaluation

The figures above show in-sample fits for the history and a single nowcast at the end. To document out-of-sample performance in the way a desk would experience it, we ran the operational workflow forward in time for both countries: at each of the last 8 quarters with an observed outcome (2024Q2 to 2026Q1), the GDP value of that quarter was masked, every later row was dropped, and the package produced its nowcast for the quarter. At the earliest origin the full tuning pass was run; at every later origin the stored tuning results were reused through `tuning = FALSE` with the differencing decisions replayed, exactly as a scheduled refresh would. Two points must be kept in mind when reading the results. The data are the final 2026Q2 vintage, not the vintages available at the time, so the exercise is pseudo-real-time: it reproduces the information set in terms of which quarters were observed, not revisions or release calendars. And eight origins are few, so the accuracy and coverage estimates are coarse; they are meant to show what the diagnostic looks like and what it reveals, and the package's refresh mode makes a longer exercise affordable

Table 6: Rolling-origin backtest: root mean squared error of the nowcast (percentage points)

Country	Origins	Best model	Ensemble	ARIMAX (suite)	AR, univariate	Best / AR	Ensemble / AR
Tunisia	8	0.83	0.71	4.32	1.41	0.59	0.51
Uganda	8	1.59	2.15	8.22	1.83	0.87	1.17

Notes: Eight origins, 2024Q2 to 2026Q1; at each origin the target of that quarter is masked and all later rows dropped. "Best model" is the model selected by cross-validated RMSE at that origin, "Ensemble" the inverse-loss-weighted average of the nine models, "ARIMAX (suite)" the suite's own ARIMA-with-regressors engine, and "AR, univariate" an `auto.arima` (BIC, non-seasonal) fitted to the target history alone, the benchmark against which the last two columns are ratios. Final-vintage data. Source: authors' calculations.

Table 7: Rolling-origin backtest: empirical coverage of the conformal prediction intervals (%)

Country	Nominal level	Best model	Ensemble	Median ensemble width (pp)
Tunisia	80%	88	100	5.5
	90%	100	100	50.8
	95%	100	100	351.8
	99%	100	100	∞
Uganda	80%	100	88	7.6
	90%	100	100	12.1
	95%	100	100	22.3
	99%	100	100	∞

Notes: Share of the eight origins at which the realised GDP growth fell inside the interval of the stated nominal level, for the best model's additive interval and the ensemble's multiplicative interval, and the median width of the ensemble interval across origins in percentage points. The 99% interval is infinite by construction with fewer than 99 calibration scores (Section 2.5). Source: authors' calculations.

(about seven minutes per origin, Section 7.2).[8]

Table 6 reports root mean squared errors against a univariate autoregressive benchmark fitted to the GDP history alone (the natural no-indicator alternative; the suite's own ARIMAX is also shown and is much worse, because fourteen exogenous regressors on fifty observations are too many for it). The results are mixed in an instructive way. For Tunisia both the best model and the ensemble beat the univariate benchmark by a wide margin (ratios of 0.83/1.41 and 0.71/1.41), and the ensemble is the more accurate of the two. For Uganda the best model beats the benchmark (1.59 against 1.83) but the ensemble does not (2.15): one origin, 2025Q3, where the ensemble's point nowcast was 9.4% against an outturn of 4.9%, accounts for most of the difference. The lesson is the one the MCS already taught: in a fifty-quarter panel the ranking of best model against ensemble is not stable, and a desk should look at both and at the suite's disagreement before choosing.

Table 7 reports empirical coverage. At the 90% and 95% levels every realised value fell inside both intervals in both countries, and at 80% coverage was 88% or 100%; with eight origins these shares cannot distinguish nominal from conservative coverage, but they rule out the under-coverage that would be the serious failure. The widths tell the more useful story. Uganda's ensemble intervals are of a size a desk can use (a median of 7.6 points at 80% and 12 points at 90%, against a target whose quarterly range over the sample is 19 points); Tunisia's are not at 90% and above, for the reason given in Section 5: in Tunisia the coverage comes at the

[8] A systematic evaluation of the same workflow across about sixty countries, against the historical mean and the last observation as benchmarks, is the subject of Abdel-Latif et al. (2026); here we document the diagnostic on the two panels of this paper.

Table 8: Leave-future-out coverage of the conformal intervals, 2022Q1 to 2026Q1

		Best model			Ensemble		
Country	Nominal level	Coverage	Median width	Infinite	Coverage	Median width	Infinite
Uganda	80%	100	11.0	0	94	10.0	0
	90%	100	15.5	0	100	13.0	0
	95%	100	20.3	24	100	17.6	24
Tunisia	80%	88	5.8	0	82	3.2	0
	90%	100	10.5	0	100	35.6	0
	95%	100	125.7	0	100	37.2	0

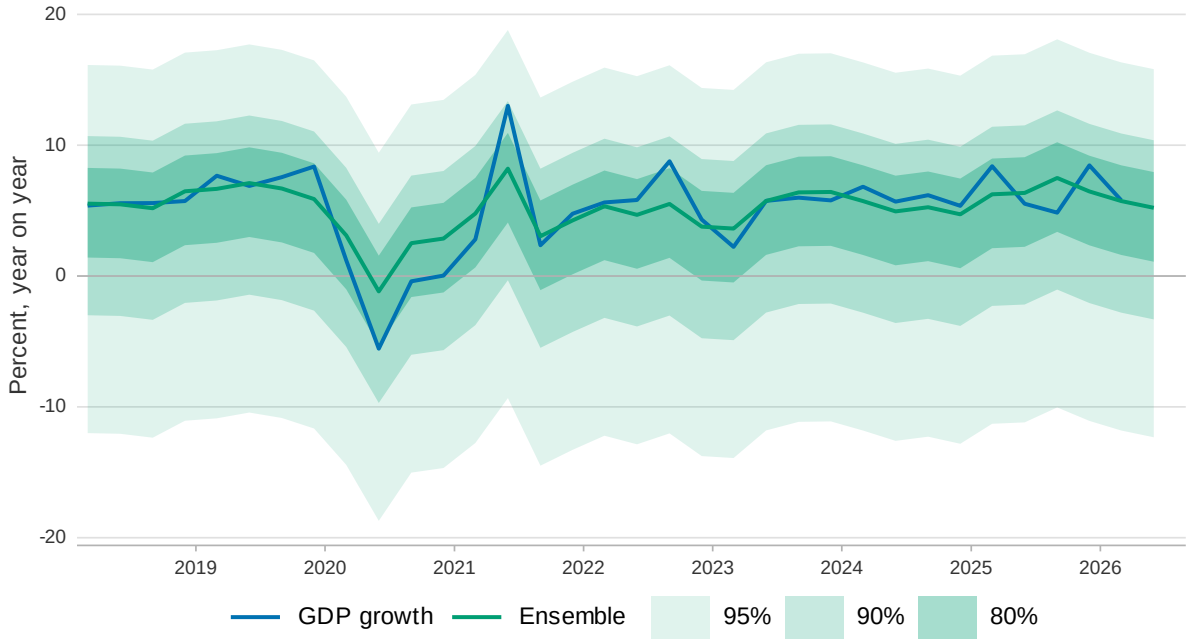
Notes: For each calibration date the band is built from the absolute out-of-sample errors of the earlier calibration dates only, under the package's rule (the $\lceil \ell(n+1) \rceil$-th smallest score, infinite when that index exceeds n), and checked against the realised value; 17 dates per country. "Best model" is the model selected on the full sample (RF for Uganda, PLS for Tunisia) with its additive interval; the ensemble is scored with the same additive rule for comparability (its published interval is the multiplicative one of AppendixB). Coverage in percent, counting an infinite band as covering; median width over the finite bands in percentage points; "Infinite" is the share of dates at which the band was unbounded. Source: authors' calculations from the package's out-of-sample record.

cost of intervals too wide to be informative.

The expanding-window record that calibrates the intervals (AppendixB) allows a finer check: at each of the 17 calibration dates from 2022Q1 to 2026Q1, the package's own quantile rule applied to the absolute out-of-sample errors of the earlier dates gives a band for that date, which is then checked against the outturn. Table 8 reports the result for the best model's additive interval and, on the same additive footing, for the ensemble. Coverage is at or above the nominal level at every level in both countries. At 80% it is close to nominal in Tunisia (88% for the best model, 82% for the ensemble) and conservative in Uganda (100% and 94%); at 90 and 95% every band covered in both countries, and the over-coverage is the pattern of Section 5: the pandemic errors sit in the calibration pool for every later date, so the bands are conservative and, at 95%, wide or, early in the window when fewer than 19 scores are available, infinite. This is the behaviour the construction is designed to have in short samples; a user who wants narrower bands at the cost of the guarantee has the `conf_method` and `top_k` arguments and the choice of calibration sample in hand.

A first real-time check became available after the fits of this paper were made. The statistical offices have since published the quarter that was nowcast: Uganda's year-on-year growth for April to June 2026 (the fourth quarter of fiscal year 2025/26) was 6.8%, against nowcasts of 5.51% for the best model and 5.21% for the ensemble (errors of −1.3 and −1.6 points), and Tunisia's preliminary estimate was 2.3%, against 2.64% and 2.38% (+0.3 and +0.1 points). Both outturns fell inside the 80% intervals. One quarter is one observation per country; the point of recording it is that these nowcasts were made and published before the outturns, which the backtest above cannot claim.

Figure 8: Uganda: fan chart of the ensemble nowcast



Notes: Actual year-on-year real GDP growth and the ensemble nowcast, 2018Q1 to 2026Q2, with the 80%, 90% and 95% expanding-window conformal prediction intervals of the ensemble (darkest to lightest). Widths at 2026Q2 are 6.9, 13.7 and 28.1 percentage points. The 99% interval is infinite with 41 calibration scores and is not drawn. Source: authors' calculations.

7. Using the package on a desk

The two subsections that follow take the outputs of the previous sections to the point of use: how the intervals are presented to a policy audience, and how the fitted models are refreshed when new data arrive within the quarter.

7.1. Communicating uncertainty: multi-level prediction intervals

The multi-level conformal intervals discussed in Section 2 are the package's response to the problem of communicating uncertainty to non-technical audiences. Because all nominal levels are computed in a single pass, fan charts can be assembled directly from `all_models_nowcast` without re-running the analysis. Figure 8 shows the fan chart for the Uganda ensemble nowcast at the 80%, 90%, and 95% levels; the 99% band is not drawn because it is infinite with 41 calibration scores, a consequence of the finite-sample quantile rule explained in Section 2.5 that users of short panels should expect. This presentation is standard in central-bank and IMF publications and provides an intuitive summary of forecast uncertainty for policy audiences.

The widths grow quickly with the level (6.9, 13.7 and 28.1 percentage points at 2026Q2), and in Tunisia they explode (4.2, 29.1 and 184.7 points). This is the signature of a short calibration sample that contains the pandemic. The 95% bound is the second-largest of 41 conformity scores, and the largest scores in both

countries are the quarters of 2020 and 2021, when every model missed the collapse and the rebound by amounts that have no precedent in the rest of the sample; the 80% bound, by contrast, is set by ordinary quarters. Distribution-free calibration cannot discount those episodes on its own, and the analyst has to decide whether they belong in the calibration history for the quarter being nowcast. The package offers the pieces for that decision (the out-of-sample path of Figure 2, the per-model intervals, the MCS-restricted ensemble, and the choice of sample) rather than a rule.

For users who want to validate empirical interval coverage on their own data, the `train_test_split` helper accepts either a proportion in $(0, 1)$, an integer number of holdout periods, or a `Date` cutoff, and partitions the data while respecting the temporal ordering. Combined with the multi-level intervals in `all_models_nowcast` and the refresh mode of Section 7.2, this is enough to run the rolling-origin diagnostic of Section 6 on any panel without bespoke scaffolding.

7.2. Reusing tuned models in production

The operational refresh problem from Section 2 is the design rationale for the `tuning` and `grid_results` arguments of `macroNowcast`. In a routine surveillance setting, a country desk runs the full tuning pipeline once per quarter (or per data revision), saves the returned `all_results` object, and uses it as the input to subsequent within-quarter refreshes.

When `tuning = FALSE`, the function reuses the best hyperparameters from the previous tuning run and refits each finalized workflow on the current data, evaluated on the same block-bootstrap CV plan. Hyperparameter search is bypassed, but `cv_param` must still be supplied because the resampling scheme is used to re-rank the models on the latest data, and the `differencing` table of the tuning run should be passed as `difference` so that the refresh applies the same transformations. Point forecasts, prediction intervals, SHAP decompositions, and model rankings are all recomputed. In practice, this collapses the cost of a refresh to model fitting on a few dozen observations, which is fast enough to embed in a scheduled script that ingests the latest data release and distributes a report.

Table 9 reports wall-clock timings for the two examples on the host used for this paper (four cores and 15 GB of memory, R 4.3.3, sequential execution), running the default nine-model suite at the settings used throughout (`grid_levels = 25`, 49 block-bootstrap replications). The full tuning pass takes about two hours per country; a refresh with `tuning = FALSE`, averaged over the seven refresh origins of the backtest, takes seven minutes, a factor of sixteen. Tuning and the calibration windows also run in parallel under a user-supplied `future` plan, which reduces the tuning pass roughly in proportion to the number of workers; we report sequential timings so that they are comparable across countries. The practical implication is that an operational pipeline can afford to retune once per release vintage and refresh several times within a quarter at negligible cost.

Table 9: Wall-clock cost of the full tuning pass and of a refresh

Country	Full tuning pass (min)	Refresh, `tuning = FALSE` (min)
Uganda	119.9	7.6
Tunisia	116.6	6.5

Notes: Minutes of wall-clock time on a four-core PC, R 4.3.3, `IMFNow` 0.0.7.1, sequential execution, nine models, 25 candidate configurations per model, 49 block-bootstrap replications, intervals and SHAP included. The refresh column is the mean over the seven `tuning = FALSE` origins of the backtest of Section 6. Source: authors' calculations.

8. Conclusions

This paper has described `IMFNow`, an R package designed around the forecasting problems of nowcasting in low-income and fragile economies. Those problems are short and noisy samples, ragged-edge releases, non-stationary regressors, severe model-selection uncertainty, the need for honest uncertainty quantification under model uncertainty, the demand for interpretable attributions for non-technical audiences, and the operational pressure of weekly within-quarter refreshes. Each constraint has shaped a specific component of the package: block-bootstrap cross-validation, the trailing-NA convention, automatic stationarity testing with back-transformation and an analyst override, the ensemble-plus-MCS treatment of model-selection uncertainty with the MCS computed on out-of-sample losses, expanding-window conformal intervals with PCS-UQ multiplicative calibration at multiple confidence levels, integrated SHAP decomposition, and the `tuning = FALSE` fast-update mode. Mixed-frequency indicators need no dedicated component: higher-frequency series enter through the ordinary regressor interface, whether aggregated to the quarterly target or left disaggregated and entered as individual high-frequency lags in the spirit of unrestricted MIDAS. The union of these components is exposed through two user-facing functions, `macroNowcast` and `outputNowcast`. The two applications and the pseudo-real-time evaluation show the workflow doing what it is meant to do, including where it exposes a weakness (the width of the ensemble intervals when the calibration history contains the pandemic) rather than hiding it. The statistical properties of the components at these sample sizes, and a broader empirical comparison across countries, are the subject of the companion methodological paper (Abdel-Latif et al., 2026).

The design transfers to other surveillance settings with the same data constraints, and the argument structure (from the forecasting problem to the API surface, rather than the other way around) applies equally to authors of similar libraries. The package is under active development. The main planned extension is panel nowcasting. Current practice at the Fund is country-by-country estimation, with regional aggregates formed by weighting the country nowcasts; the direction we intend to take is partial pooling, in which hyperparameters and ensemble weights are shared across countries with similar data environments while the model coefficients remain country-specific, so that the countries with the shortest samples borrow the tuning, not the parameters, of their neighbours. The data-planning implication, harmonised predictor definitions across countries, is already shaping the IMF's nowcasting data vintages. A second extension is tighter

integration with real-time data pipelines, so that the vintage structure of the data, which the pseudo-real-time exercise of this paper could not reproduce, becomes part of the evaluation.

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AppendixA. Block-bootstrap cross-validation as implemented

The motivation for this scheme and its finite-sample properties are the subject of Abdel-Latif et al. (2026); this appendix records only what the package does. `cv_plan_custom(data_train, reps, block_size, depvar)` returns an `rsample` object of `reps` training and assessment splits of the n training rows, constructed as follows.

1. *Block length.* If `block_size` is `NULL`, b is the circular-block-bootstrap length of Patton et al. (2009) computed from the dependent variable with `np::b.star`, rounded up, with a floor of 4; it is then capped at $\lfloor n/4 \rfloor$. Otherwise the user's value is used, subject to the same cap.
2. *Blocks.* One block of length b starts at every observation $1, \ldots, n$; blocks that run past n wrap around to the start of the series (circular construction), so that end-of-sample observations appear in as many blocks as any other.
3. *Assessment window.* For each replication one block is drawn at random and its first $a = \min(b, 4)$ observations form the held-out assessment window.
4. *Leakage control.* Every block that contains any observation of the assessment window is removed from the sampling pool.
5. *Training set.* Blocks are drawn with replacement from the remaining pool and concatenated until $n - a$ training rows are obtained.
6. *Seeding.* The plan is generated under the run's `seed`, so that a refresh with `tuning = FALSE` re-ranks the models on the same splits as the tuning run.

In the two panels of this paper ($n = 53$ observed quarters) the estimator selects $b = 18$ for Uganda, capped to 13 by the $n/4$ rule, and $b = 4$ for Tunisia; each replication holds out four consecutive quarters and trains on 49 rows drawn from the admissible blocks.

AppendixB. Conformal calibration as implemented

As for AppendixA, the coverage properties of this construction are studied in Abdel-Latif et al. (2026); this appendix records the construction as implemented in version 0.0.7.1. Let M be the number of models, $y_1, \ldots, y_n$ the observed (possibly differenced) target, and θ^*_m the configuration of model m selected on the full-sample tuning results. With `conf_method = "jackknife"`:

1. *Windows.* For $t = n_0 + 1, \ldots, n$ with $n_0 = \max(10, M + 2)$ ($n_0 = 11$ for the nine-model suite), every model's selected configuration is refitted on observations $1, \ldots, t-1$ and predicts y_t; write $\hat{y}_{m,t}$. In addition, the top-K configurations of every model by cross-validated loss ($K = 10$) are refitted on the same window, giving a pseudo-population $\mathcal{P}_t$ of up to MK predictions of y_t. (Configurations are selected once, on the

full sample, and refitted in each window, not re-selected; the usual finite-sample conformal coverage guarantee therefore does not apply exactly.)

2. *Per-model scores.* For each model, $s_{m,t} = |y_t - \hat{y}_{m,t}|$ at every window in which its prediction is finite. Models with fewer than five scores receive `NA` intervals.
3. *Ensemble prediction.* At every window in which all M predictions are finite, $\hat{y}^E_t = \sum_m w_m \hat{y}_{m,t}$ with the inverse-CV-loss weights w_m of the ensemble.
4. *Studentised ensemble scores.* With c_t the median of $\mathcal{P}_t$ and, for the run's level $1-\alpha$, the half-widths $h^-_t = \max\{c_t - q_{\alpha/2}(\mathcal{P}_t), f\}$ and $h^+_t = \max\{q_{1-\alpha/2}(\mathcal{P}_t) - c_t, f\}$, where f is one per cent of the standard deviation of the target (a floor that prevents division by zero when the candidates agree), the score is $\gamma_t = (c_t - y_t)/h^-_t$ if $y_t \le c_t$ and $(y_t - c_t)/h^+_t$ otherwise.
5. *Quantiles.* For a reported level ℓ and n_s scores, the critical value is the $\lceil \ell(n_s+1) \rceil$-th smallest score, or $+\infty$ when that index exceeds n_s.
6. *Per-model intervals* (additive): $\hat{y}_{m,n+1} \pm s^*_m(\ell)$, where $\hat{y}_{m,n+1}$ is the prediction of the model fitted on the full sample and $s^*_m(\ell)$ the critical value of its own scores. With `individual_multiplicative = TRUE` the per-model interval is instead built as in step 7 from the model's own top-K pseudo-population.
7. *Ensemble interval* (multiplicative): with $\mathcal{P}_{n+1}$ the pseudo-population of full-sample fits at the nowcast row, c its median, $h^-(\ell)$ and $h^+(\ell)$ its half-widths at level ℓ as in step 4, and $\gamma^*(\ell)$ the critical value of the γ_t, the interval is $[\hat{y}^E_{n+1} - \gamma^*(\ell)\, h^-(\ell),\ \hat{y}^E_{n+1} + \gamma^*(\ell)\, h^+(\ell)]$. The interval therefore widens where the candidate configurations disagree at the nowcast row and narrows where they agree.
8. *Back-transformation.* If the target was differenced, every bound is converted to levels by adding the last observed level.
9. *Out-of-sample record.* The $\hat{y}_{m,t}$ and $\hat{y}^E_t$ are returned in `nowcast_oos` and are the inputs of the Model Confidence Set.

With `conf_method = "split"`, the same scores are computed once on the second temporal half of the sample, with the models fitted on the first half. In the examples of this paper the calibration windows run from 2016Q1 to 2026Q1 (41 dates); the ensemble scores number 32 for Uganda and 36 for Tunisia because the random forest (whose selected block length exceeds the shortest windows) and the ARIMAX did not produce predictions in the earliest windows.

AppendixC. Hyperparameter search space

Table C.10 lists the parameters tuned for each default model and their ranges. For each model, `grid_levels` candidate configurations (25 in this paper) are drawn from a space-filling design over these ranges, every candidate is evaluated on every block-bootstrap resample, and the configuration with the best mean `metric_select` is refitted on the full sample.

Table C.10: Tuned hyperparameters of the default model suite and their ranges

Model	Engine	Tuned parameters (range)	Fixed settings
Elast	glmnet	penalty $\lambda \in [10^{-10}, 1]$ (log scale); mixture $\alpha \in [0.05, 1]$	
PLS	mixOmics	components $\in \{1, \ldots, 4\}$; predictor proportion $\in [0, 1]$	
RF	rangerts	variables per split $\in \{1, \ldots, 15\}$; block length $\in \{10, \ldots, 30\}$	1,000 trees
XGboost	xgboost	tree depth $\in \{1, \ldots, 15\}$; variables per split $\in \{1, \ldots, J\}$	1,000 trees, learning rate 0.05, subsample 0.5
BART	dbarts	trees $\in \{50, \ldots, 500\}$; prior coefficient $\in [0, 1]$; prior exponent $\in [1, 3]$	1,000 burn-in draws
Cubist	Cubist	committees $\in \{1, \ldots, 100\}$; neighbours $\in \{2, \ldots, 9\}$; max rules $\in \{1, \ldots, 500\}$	
NNETAR	nnetar	AR inputs $p \in \{0, \ldots, 5\}$ (0 if `lags_y > 0`); hidden units $\in \{1, \ldots, 10\}$	20 networks, decay 0.1, 100 epochs
AR	auto_arima	none (orders selected by BIC)	non-seasonal, predictors as regressors
DFM	dfms	factors $r \in \{1, \ldots, 5\}$; VAR lags $p \in \{1, \ldots, 4\}$	EM tolerance 10^{-5}, 2,500 iterations

Notes: J is the number of predictors. Ranges are the `dials` defaults except where the package narrows them (Cubist neighbours, NNETAR with user lags). The ML models share a recipe that imputes, centres and scales the predictors; the time-series engines retain the date column.

AppendixD. Predictors

Tables D.11 and D.12 list the predictors of the two panels with their source, native frequency, the transformation applied in the data vintage, and the differencing decision taken by the KPSS step of `macroNowcast` (the `differencing` element of the fitted object). The vintage is the quarterly year-on-year stream of NowHub's data-collection pipeline (2026Q2, built on 12 August 2026): monthly series are aggregated to calendar quarters (averages for indices, sums for flows, end-of-period values for stocks) and then transformed as stated in the table; satellite series are country means of daily or multi-day retrievals aggregated to months first. Quarterly GDP enters with a two-month publication shift and monthly Haver series for Tunisia and the China PMI with a one-month shift, so that each quarter's row holds the information available in the quarter in which it is published. Both panels run from 2013Q1 to 2026Q2 with no missing values in the predictors; the target is observed through 2026Q1 and left in levels by the test in both countries. The metadata file distributed with the paper (`dat/predictor_metadata.csv`) carries the provider identifiers. Table D.13 shows the last five quarters of the Tunisia panel.

AppendixE. Cross-validated and out-of-sample accuracy by model

Tables E.14 and E.15 compare, for every model, the cross-validated RMSE that drives selection and the ensemble weights with the RMSE of the expanding-window out-of-sample predictions over the calibration dates (AppendixB), both over the full calibration window and excluding the eight pandemic quarters 2020Q1 to 2021Q4. Table E.16 is the Tunisia Model Confidence Set.

Table D.11: Uganda: predictors and the differencing decision of the KPSS step

Predictor	Column	Source	Native frequency	Transformation in the vintage	KPSS diff.
Composite index of economic activity	ciea	Bank of Uganda (disseminated indicators)	Monthly	Year-on-year % change	no
Business indicator	business_indicator_uga	Bank of Uganda (disseminated indicators)	Monthly	Level	no
Domestic credit	domestic_credit	Bank of Uganda (disseminated indicators)	Monthly (end of period)	Year-on-year % change	no
Exports	exports	Bank of Uganda (disseminated indicators)	Monthly	Year-on-year % change	yes
Real effective exchange rate	reer	IMF (iData effective exchange rates)	Monthly	Year-on-year % change	yes
Terms of trade	trade_terms	Bank of Uganda (disseminated indicators)	Monthly	Year-on-year % change	no
PMI, China	pmi_CHN	S&P Global / Markit via Haver Analytics	Monthly	Year-on-year % change	no
World Uncertainty Index	wui	Ahir, Bloom and Furceri (EIU country reports)	Quarterly	Level	no
Google Trends, tourism	gti_tourism	Google Trends (web search)	Monthly	Level	yes
Aerosol optical depth	mean_aod_55	NASA MODIS (satellite)	Daily to monthly	Year-on-year log difference	no
NDVI anomaly	mean_ndvi_anomaly	FAO / NOAA agricultural stress indicators (satellite)	10-day to monthly	Level	no
NDWI	mean_ndwi	NASA MODIS (satellite)	5 to 16 days to monthly	Level	yes
NO2 column density	mean_no2	ESA Sentinel-5P TROPOMI / NASA Aura OMI (satellite)	Daily to monthly	Year-on-year log difference	no
Night-time lights	mean_ntl	NASA VIIRS Black Marble VNP46A2 (satellite)	Daily to monthly	Year-on-year log difference	no

Notes: Source, native frequency and transformation as recorded in the registries of the IMF nowcasting data pipeline for vintage 2026Q2. "KPSS diff." is `yes` when the series entered the models in first differences because the KPSS test embedded in `forecast::auto.arima` selected $d = 1$ (applied on top of the vintage transformation). Satellite series are country means.

Table D.12: Tunisia: predictors and the differencing decision of the KPSS step

Predictor	Column	Source	Native frequency	Transformation in the vintage	KPSS diff.
Imports	`imports_mil_tnd`	National Institute of Statistics (INS) via Haver Analytics	Monthly	Year-on-year % change	no
Broad money (M2)	`m2_eop_sa_thous_tnd`	Central Bank of Tunisia via Haver Analytics	Monthly (end of period)	Year-on-year % change	no
NO2 column density	`mean_no2`	ESA Sentinel-5P TROPOMI / NASA Aura OMI (satellite)	Daily to monthly	Year-on-year log difference	no
Travellers	`travelers_total_thous`	National authorities via Haver Analytics	Monthly	Year-on-year % change	no
Industrial production	`ipi_2010_100_nsa`	National Institute of Statistics (INS) via Haver Analytics	Monthly	Year-on-year % change	no
Real effective exchange rate	`reer`	IMF (iData effective exchange rates)	Monthly	Year-on-year % change	yes
NDVI anomaly	`mean_ndvi_anomaly`	FAO / NOAA agricultural stress indicators (satellite)	10-day to monthly	Level	no
Night-time lights	`mean_ntl`	NASA VIIRS Black Marble VNP46A2 (satellite)	Daily to monthly	Year-on-year log difference	no
Aerosol optical depth	`mean_aod_55`	NASA MODIS (satellite)	Daily to monthly	Year-on-year log difference	no
NDWI	`mean_ndwi`	NASA MODIS (satellite)	5 to 16 days to monthly	Level	yes
Google Trends, tourism	`gti_tourism`	Google Trends (web search)	Monthly	Level	yes
World Uncertainty Index	`wui`	Ahir, Bloom and Furceri (EIU country reports)	Quarterly	Level	yes
Exports	`exports_mil_tnd`	National Institute of Statistics (INS) via Haver Analytics	Monthly	Year-on-year % change	no
Oil price	`poil`	Haver Analytics (world commodity prices)	Monthly	Year-on-year log difference	no

Notes: As Table D.11.

Table D.13: Last five quarters of the Tunisia panel passed to `macroNowcast`

Variable	2025Q2	2025Q3	2025Q4	2026Q1	2026Q2
Real GDP growth, y/y (target)	3.20	2.40	2.72	2.56	`NA`
Imports	0.03	0.08	0.06	0.06	0.21
Broad money (M2)	0.10	0.11	0.10	0.11	0.10
NO2 column density	-0.29	0.14	-0.04	0.02	0.47
Travellers	0.19	0.05	0.14	-0.07	-0.16
Industrial production	0.03	0.02	0.05	0.04	0.00
Real effective exchange rate	0.04	0.04	0.02	0.02	0.02
NDVI anomaly	0.33	0.23	0.29	0.42	0.36
Night-time lights	0.24	1.58	-0.03	0.06	-0.05
Aerosol optical depth	-0.32	0.00	-0.38	-0.25	-0.26
NDWI	-0.98	-1.16	-0.77	-0.75	-1.09
Google Trends, tourism	11.33	50.33	70.33	65.00	57.33
World Uncertainty Index	0.39	0.40	0.33	0.28	0.28
Exports	0.05	0.01	0.10	0.06	0.12
Oil price	-0.24	-0.15	-0.18	0.06	0.35

Notes: As Table 1.

Table E.14: Uganda: cross-validated RMSE and expanding-window out-of-sample RMSE by model

Model	CV RMSE	s.e.	Calibration dates	OOS RMSE	OOS RMSE excl. 2020–21
RF	2.15	0.21	32	3.24	1.68
Elast	2.21	0.22	41	3.09	2.55
BART	2.24	0.22	41	3.38	2.48
XGboost	2.28	0.20	41	3.01	2.20
PLS	2.41	0.21	41	3.22	2.99
DFM	2.45	0.23	41	3.84	2.85
Cubist	2.52	0.19	41	3.70	3.18
NNETAR	3.49	0.27	41	3.66	3.39
AR	5.33	0.59	36	6.79	6.88
Ensemble	–	–	32	2.90	1.90

Notes: CV RMSE is the mean over the 49 block-bootstrap replications (s.e. its standard error), on which the models are ranked. OOS RMSE is computed over the calibration dates (2016Q1 to 2026Q1) at which the model, refitted on data strictly before each date, produced a prediction; the ensemble is defined only at dates where every model did. The last column drops the eight quarters 2020Q1 to 2021Q4 (24 dates remain for the ensemble). The full-window OOS errors include the pandemic quarters, which the CV resamples see in both training and assessment roles.

Table E.15: Tunisia: cross-validated RMSE and expanding-window out-of-sample RMSE by model

Model	CV RMSE	s.e.	Calibration dates	OOS RMSE	OOS RMSE excl. 2020–21
PLS	2.43	0.33	41	10.30	1.46
RF	2.49	0.44	37	4.46	0.93
BART	2.55	0.46	41	4.41	0.90
Cubist	2.58	0.42	41	3.36	0.77
XGboost	2.61	0.45	41	3.99	1.52
Elast	2.73	0.43	41	14.63	1.11
DFM	2.79	0.34	41	3.36	1.85
NNETAR	3.84	0.41	41	4.88	2.69
AR	4.56	0.52	36	18.61	1.70
Ensemble	–	–	36	4.47	0.90

Notes: As Table E.14. The out-of-sample RMSE of the two regularised linear models and of the ARIMAX over the full window is dominated by the pandemic quarters 2020Q2 and 2021Q2, in which their predictions from pre-pandemic fits were extreme (Section 5); excluding 2020Q1 to 2021Q4 (28 dates for the ensemble) their errors are of the same order as the cross-validated ones.

Table E.16: Tunisia: 90% Model Confidence Set on out-of-sample squared errors

Model	Avg. squared error	p-value, H_{0,M_k}	MCS p-value	In the 90% MCS
Elast	243.52	0.52	0.52	yes
AR	346.39	0.56	0.56	yes
PLS	120.56	0.58	0.58	yes
RF	20.49	0.46	0.58	yes
XGboost	18.01	0.31	0.58	yes
BART	22.08	0.38	0.58	yes
NNETAR	27.09	0.25	0.58	yes
Ensemble	19.97	0.61	0.61	yes
Cubist	12.77	0.99	0.99	yes
DFM	12.75	1.00	1.00	yes

Notes: As Table 2, over the 36 calibration quarters at which all ten candidates produced a prediction. No candidate is eliminated at the 90% level.

AppendixF. A structural-break check before the first run

The package does not test for structural breaks (Section 2.3). The following lines, run once on a new country panel, apply the supremum-*F* test and the Bai and Perron (2003) breakpoint estimator of `strucchange` to the mean of the target and plot the result; the same call with a predictor on the right-hand side checks the stability of a nowcasting relationship.

```
library(strucchange)
y  <- ts(na.omit(uga_data$gdp), start = c(2013, 1), frequency = 4)
fs <- Fstats(y ~ 1, from = 0.15)          # sup-F test for a mean shift
sctest(fs)                                # p-value of the sup-F test
bp <- breakpoints(y ~ 1, h = 0.15)        # Bai-Perron multiple breakpoints
summary(bp); plot(bp)                     # BIC chooses the number of breaks
```

A significant break that coincides with a known event can be handled by passing an explicit dummy regressor, by starting the sample after the break if enough observations remain, or by forcing the differencing decision through the `difference` argument when the apparent unit root is a level shift.



PUBLICATIONS

IMFNow: The IMF Nowcasting Framework in R

Working Paper No. WP/2026/240